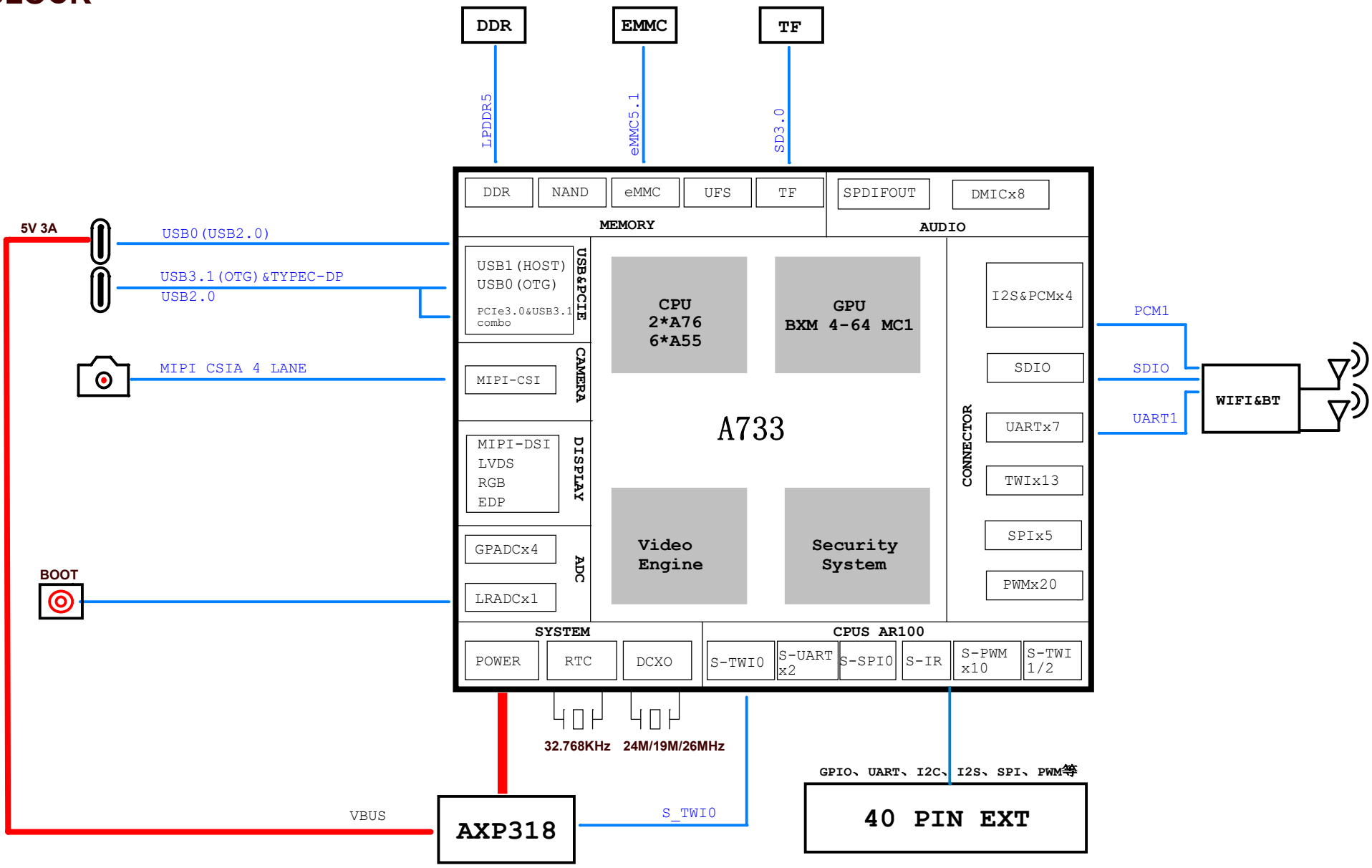
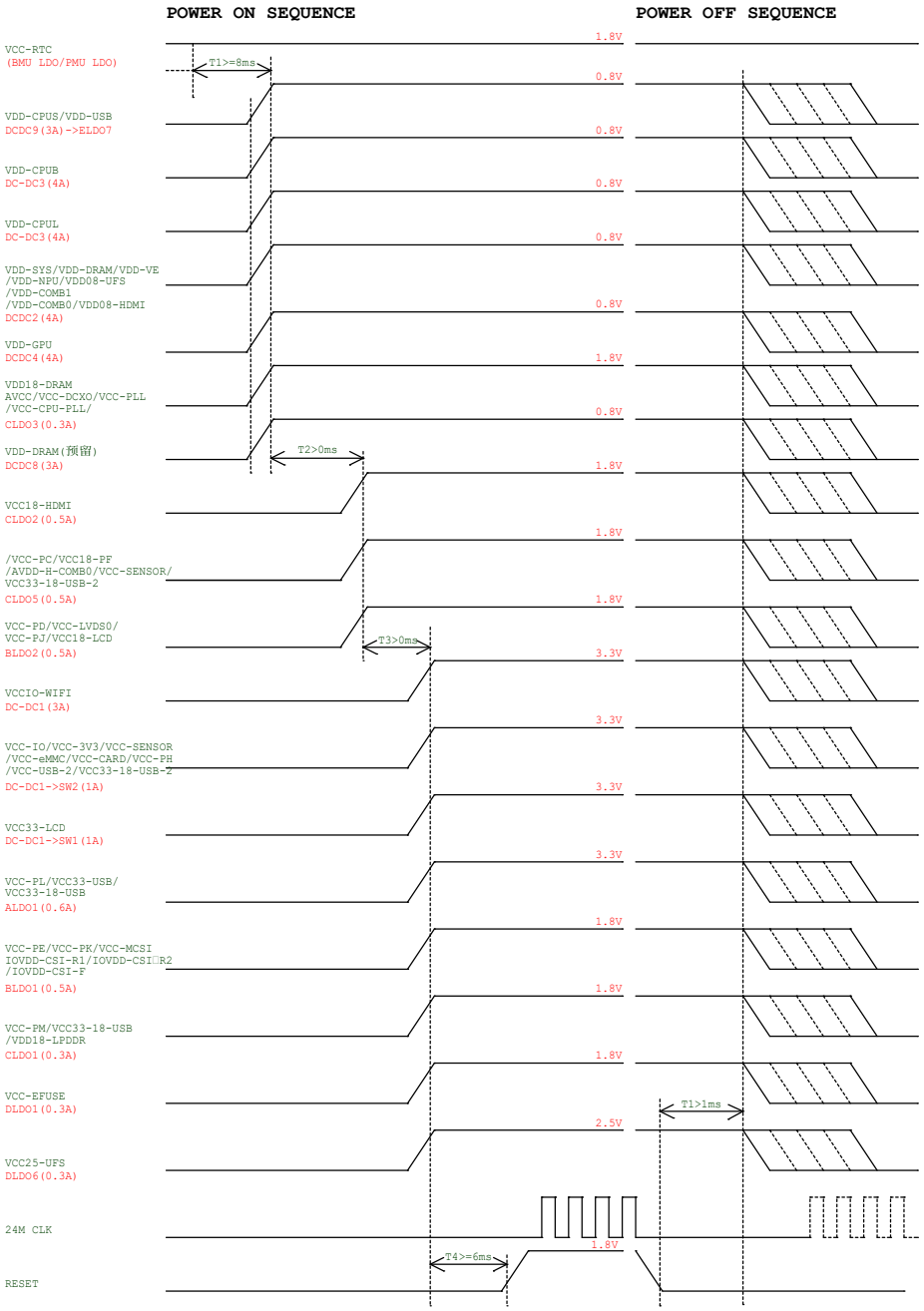
BLOCK



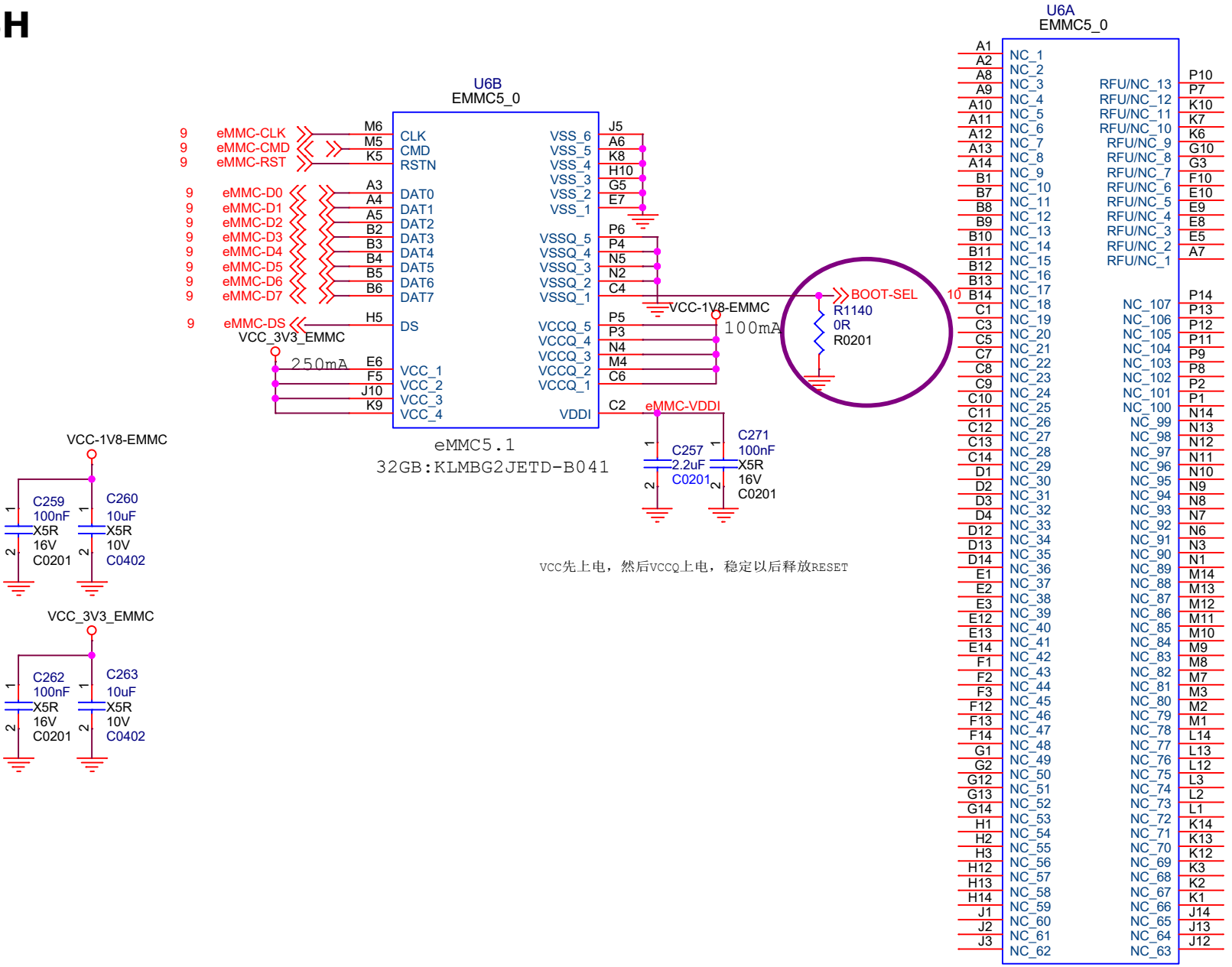
Shenzhen Xunlong Software Co., Ltd						
Schematic Name:	OPi ZERO 3W					
Page Name:	01_BLOCK DIAGRAM					
Page Size:	A3	Sheet:	1	Of:	18	Rev: V1.2

POWER SEQUENCE



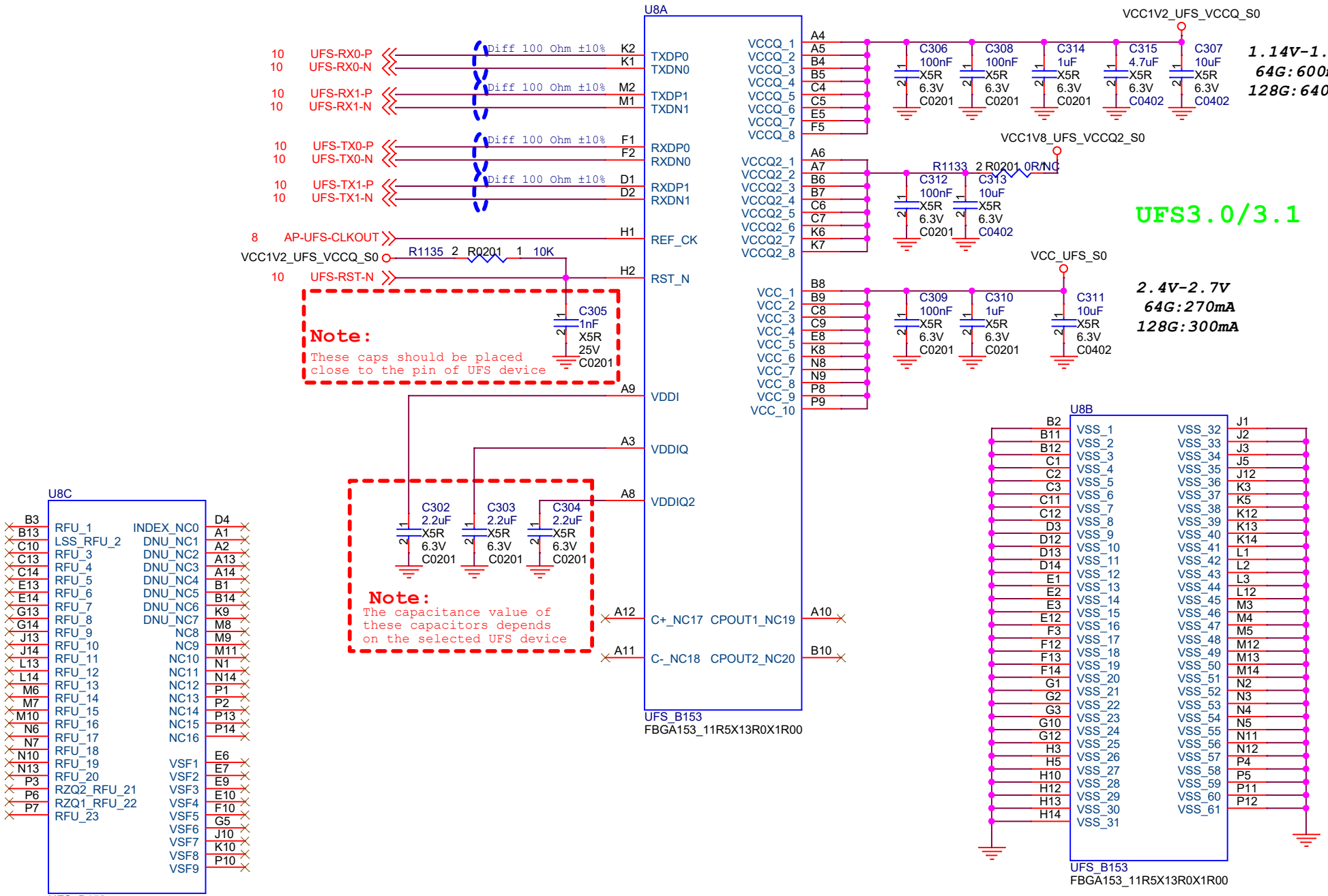
Shenzhen Xunlong Software Co., Ltd						
Schematic Name:	OPi ZERO 3W					
Page Name:	02_POWER SEQUENCE					
Page Size:	A3	Sheet:	2	Of:	18	Rev: V1.2

eMMC FLASH

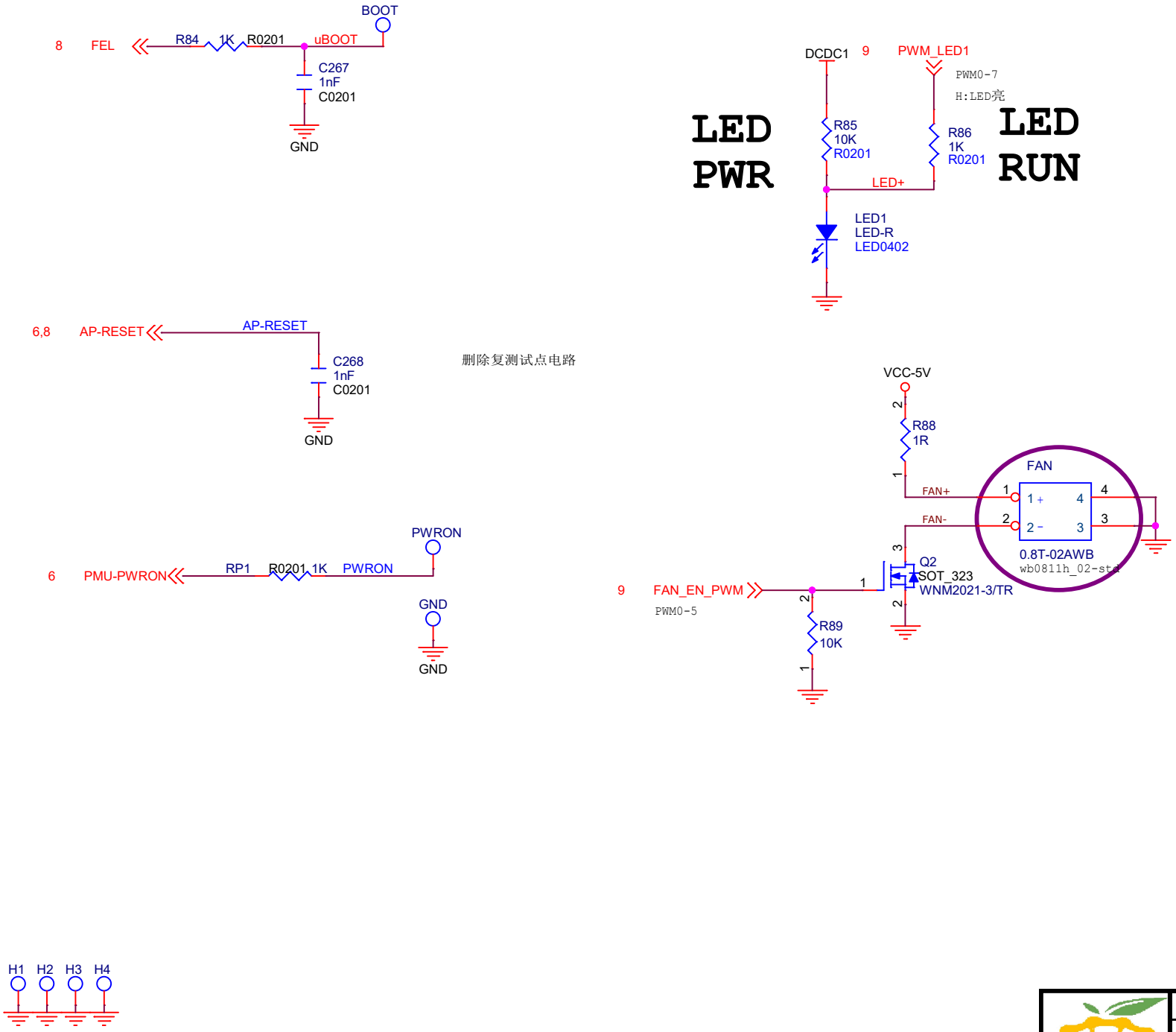


VCC先上电，然后VCCQ上电，稳定以后释放RESET

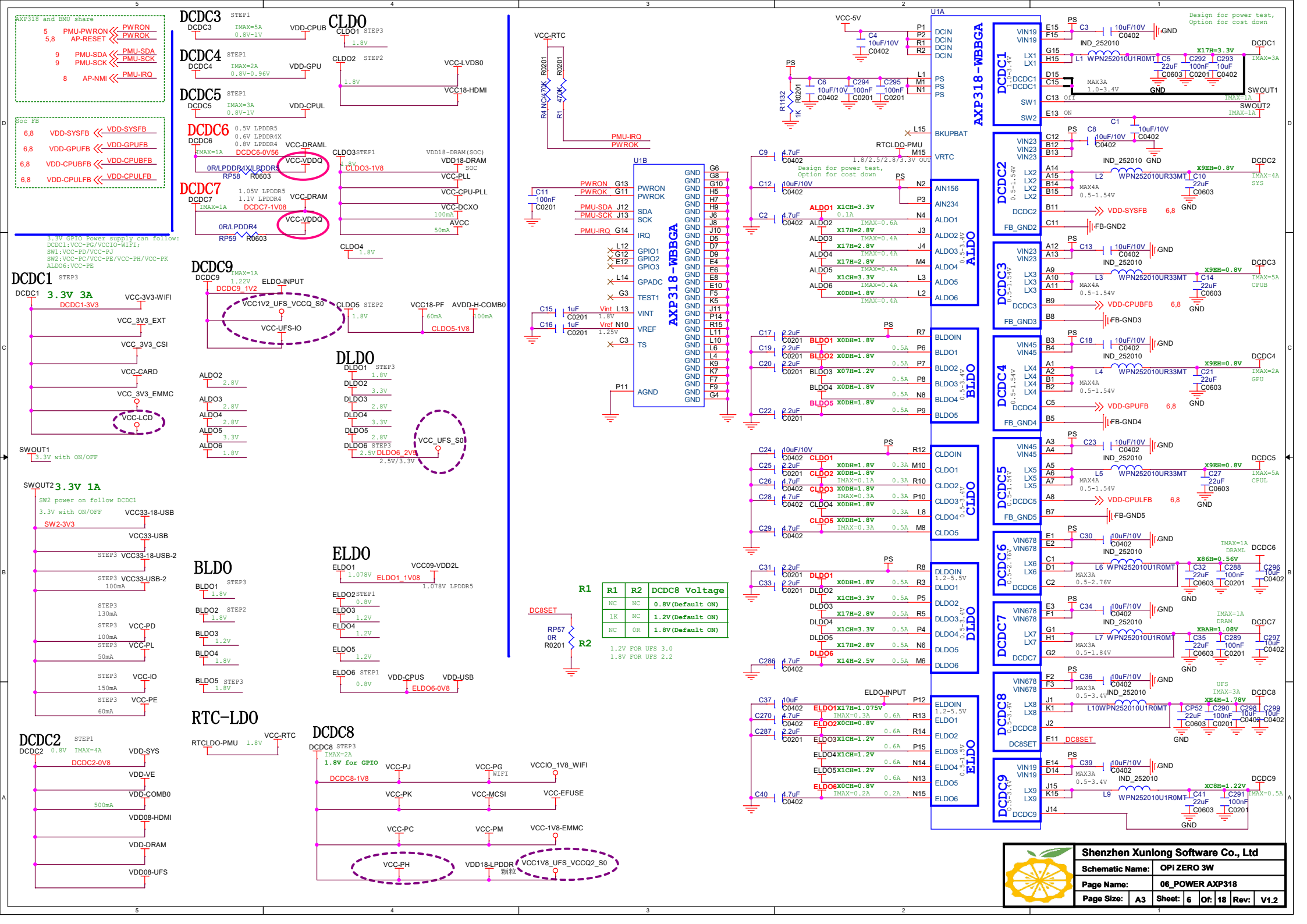
Shenzhen Xunlong Software Co., Ltd							
Schematic Name:	OPi ZERO 3W						
Page Name:	03_eMMC						
Page Size:	A3	Sheet:	3	Of:	18	Rev:	V1.2



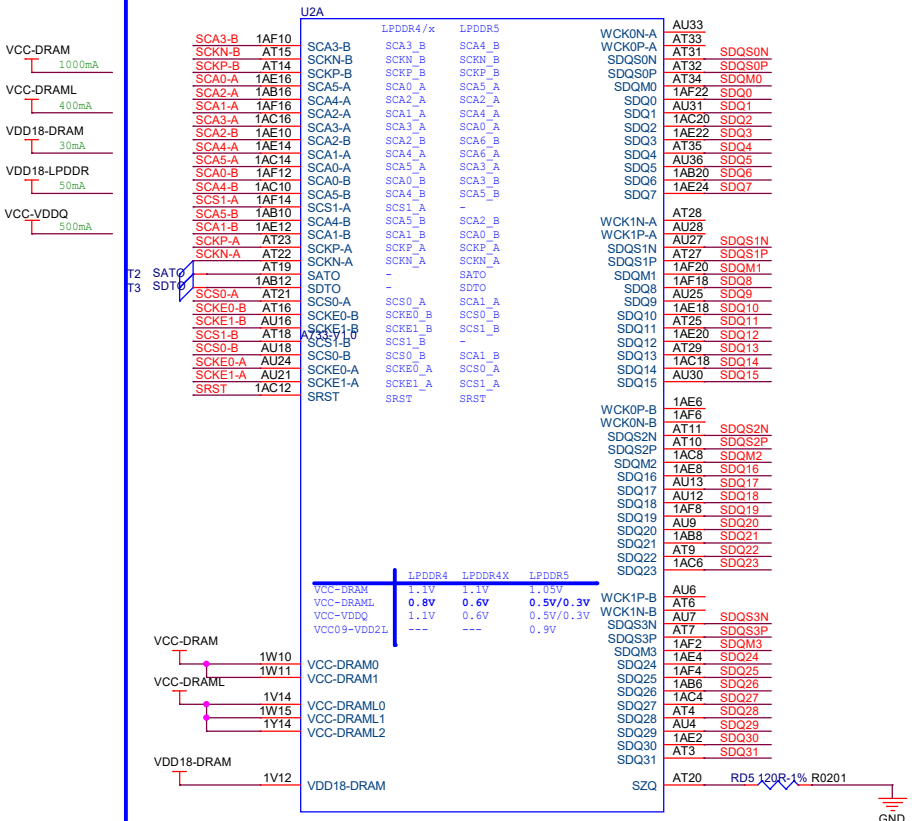
Shenzhen Xunlong Software Co., Ltd					
Schematic Name:	OPi ZERO 3W				
Page Name:	04_UFS				
Page Size:	A3	Sheet:	4	Of: 18	Rev: V1.2



Shenzhen Xunlong Software Co., Ltd							
Schematic Name:	OPi ZERO 3W						
Page Name:	05_DEBUG/TEST/LED/FAN						
Page Size:	A3	Sheet:	5	Of:	18	Rev:	V1.2

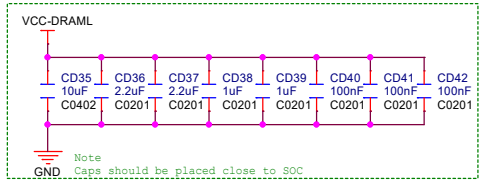

AXP318 and BMU share
PMU-PWRON
PWRON
AP-RESET
PWROK
PMU-SDA
PMU-SCK
AP-NMI
PMU-IRQ
Soc PB
VDD-SYSFB
VDD-GPUFB
VDD-CPUBFB
VDD-CPULFB
3.3V GPIO Power supply can follow:
DCDC1:VCC-PG/VCCIO-WIFI;
SW1:VCC-PD/VCC-PJ
SW2:VCC-PC/VCC-PE/VCC-PH/VCC-PK
ALDO6:VCC-PE
DCDC1
3.3V 3A
DCDC1-3V3
VCC-3V3-WIFI
VCC_3V3_EXT
VCC_3V3_CSI
VCC-CARD
VCC_3V3_EMMC
VCC-LCD
SWOUT1
3.3V with ON/OFF
SWOUT2
3.3V 1A
SW2 power on follow DCDC1
3.3V with ON/OFF
SW2-3V3
VCC33-18-USB
VCC33-USB
VCC33-18-USB-2
VCC33-USB-2
VCC-PD
VCC-PL
VCC-IO
VCC-PE
DCDC2
DCDC2-0V8
VDD-SYS
VDD-VE
VDD-COMB0
VDD08-HDMI
VDD-DRAM
VDD08-UFS
DCDC3
DCDC4
DCDC5
DCDC6
0.5V LPDDR5
0.6V LPDDR4X
0.8V LPDDR4
VCC-DRAML
DCDC6-0V56
0R/LPDDR4X/LPDDR5
RP58
R0603
DCDC7
1.05V LPDDR5
1.1V LPDDR4
VCC-DRAM
DCDC7-1V08
0R/LPDDR4
RP59
CLDO
CLDO1
CLDO2
CLDO3
CLDO4
CLDO3-1V8
VDD18-DRAM(SOC)
VDD18-DRAM
SOC
VCC-PLL
VCC-CPU-PLL
VCC-DCXO
AVCC
100mA
50mA
VCC-LVDS0
VCC18-HDMI
DCDC9
DCDC9_1V2
ELDO-INPUT
VCC1V2_UFS_VCCQ_S0
VCC-UFS-IO
CLDO5
VCC18-PF
AVDD-H-COMB0
CLDO5-1V8
DLDO
DLDO1
DLDO2
DLDO3
DLDO4
DLDO5
DLDO6
DLDO6_2V5
2.5V/3.3V
VCC_UFS_S0
ALDO2
ALDO3
ALDO4
ALDO5
ALDO6
BLDO
BLDO1
BLDO2
BLDO3
BLDO4
BLDO5
ELDO
ELDO1
ELDO1_1V08
ELDO2
ELDO3
ELDO4
ELDO5
ELDO6
VCC09-VDD2L
VDD-CPUS
VDD-USB
ELDO6-0V8
RTC-LDO
RTCLDO-PMU
VCC-RTC
DCDC8
1.8V for GPIO
DCDC8-1V8
VCC-PJ
VCC-PG
WIFI
VCCIO_1V8_WIFI
VCC-PK
VCC-MCSI
VCC-EFUSE
VCC-PC
VCC-PM
VCC-1V8-EMMC
VCC-PH
VDD18-LPDDR
颗粒
VCC1V8_UFS_VCCQ2_S0
VCC-RTC
R4 NC/470K
R1 470K
PMU-IRQ
PWROK
U1B
AXP318-WBBGA
PWRON
SDA
SCK
PMU-SDA
PMU-SCK
IRQ
GPIO1
GPIO2
GPIO3
GPADC
TEST1
VINT
VREF
TS
AGND
C11 100nF C0201
C15 1uF C0201
C16 1uF C0201
Vint 1.8V
Vref 1.25V
R1
R2
DC8SET
RP57 0R R0201
R1 R2 DCDC8 Voltage
NC NC 0.8V(Default ON)
1K NC 1.2V(Default ON)
NC 0R 1.8V(Default ON)
1.2V FOR UFS 3.0
1.8V FOR UFS 2.2
U1A
AXP318-WBBGA
DCIN
BKUPBAT
VRTC
AIN156
AIN234
ALDO1
ALDO2
ALDO3
ALDO4
ALDO5
ALDO6
BLDOIN
BLDO1
BLDO2
BLDO3
BLDO4
BLDO5
CLDOIN
CLDO1
CLDO2
CLDO3
CLDO4
CLDO5
DLDOIN
DLDO1
DLDO2
DLDO3
DLDO4
DLDO5
DLDO6
ELDOIN
ELDO1
ELDO2
ELDO3
ELDO4
ELDO5
ELDO6
VCC-5V
RTCLDO-PMU
Design for power test, Option for cost down
DCDC1
DCDC2
DCDC3
DCDC4
DCDC5
DCDC6
DCDC7
DCDC8
DCDC9
X17H=3.3V
X9EH=0.8V
X86H=0.56V
XBAH=1.08V
XE4H=1.78V
XC8H=1.22V
VDD-SYSFB
VDD-CPUBFB
VDD-GPUFB
VDD-CPUFLB
FB-GND2
FB-GND3
FB-GND4
FB-GND5
Shenzhen Xunlong Software Co., Ltd
Schematic Name: OPi ZERO 3W
Page Name: 06_POWER AXP318
Page Size: A3
Sheet: 6
Of: 18
Rev: V1.2

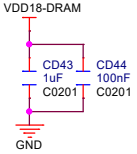
LPDDR4-4X

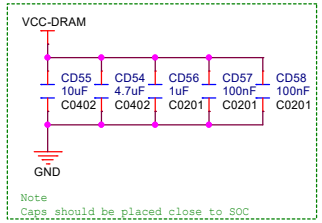


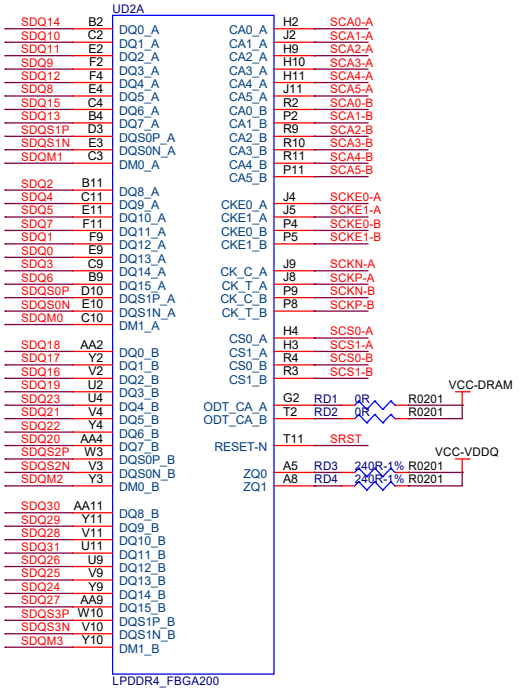
	LPDDR4	LPDDR4X	LPDDR5
VCC-DRAM	1.1V DCDC7	1.1V DCDC7	1.05V DCDC7
VCC-DRAML	0.8V DCDC6	0.6V DCDC6	0.5/0.3V DCDC6
VCC-VDDQ	1.1V DCDC7	0.6V DCDC6	0.5/0.3V DCDC6
VCC09-VDD2L	/	/	0.9V ELDO1

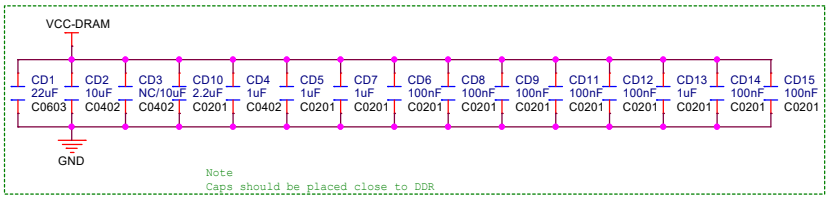
VCC-VDDQ supply is modified on the power AXP318 page

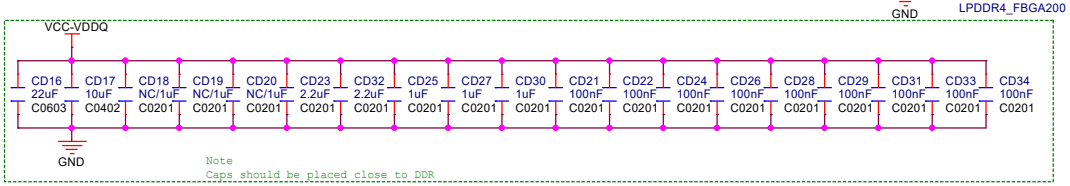


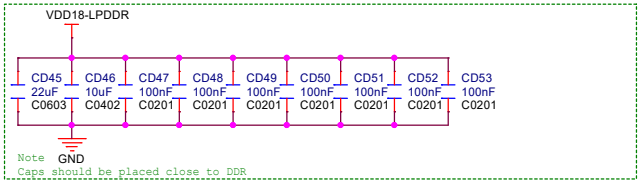


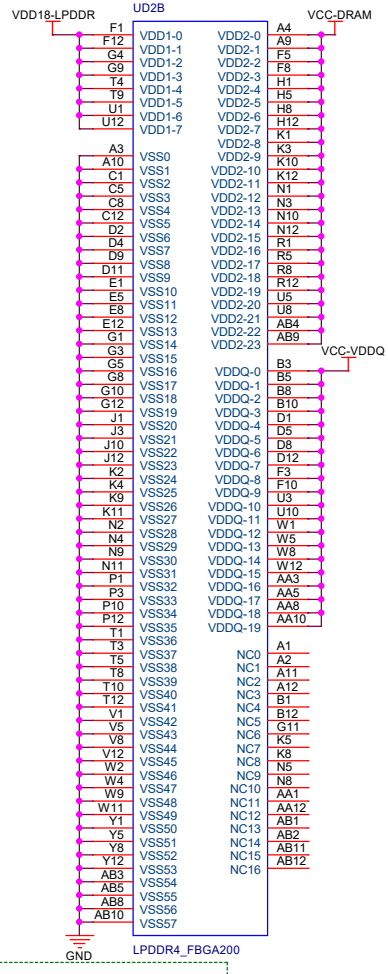






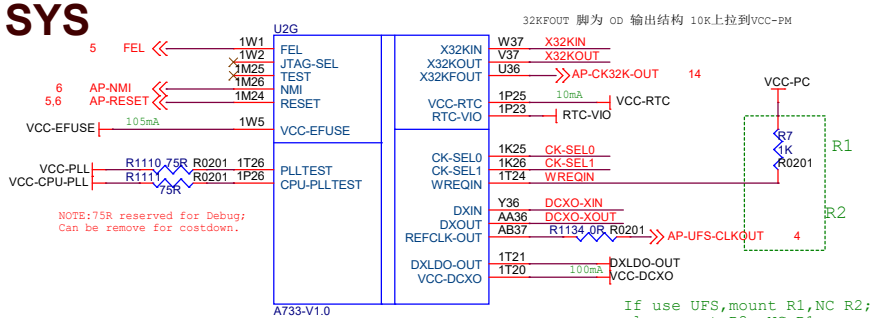


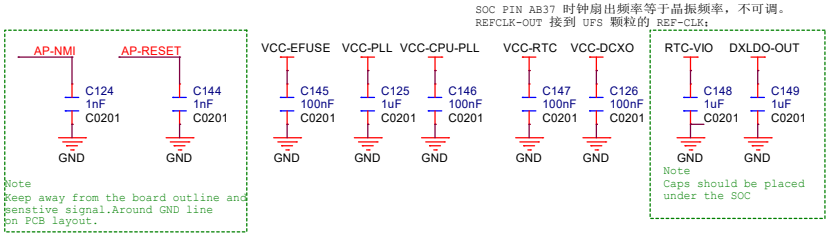




Shenzhen Xunlong Software Co., Ltd			
Schematic Name:	OPi ZERO 3W		
Page Name:	07_LPDDR4 32X1		
Page Size:	A3	Sheet: 7 Of: 18	Rev: V1.2

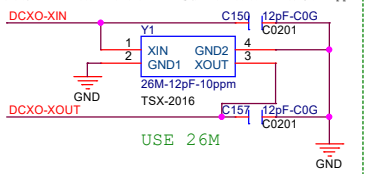
SYS





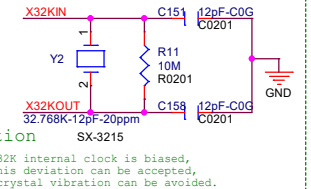
DCXO-SYS

当需要 DCXO 扇出时钟给 UFS 使用时，晶体选型频偏≤10ppm;
当不需要 DCXO 扇出时钟给 UFS 使用时，晶体选型频偏≤20ppm。



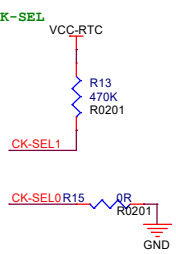
OSC-32K

外部 32.768kHz 晶振的精度要求小于≤20ppm



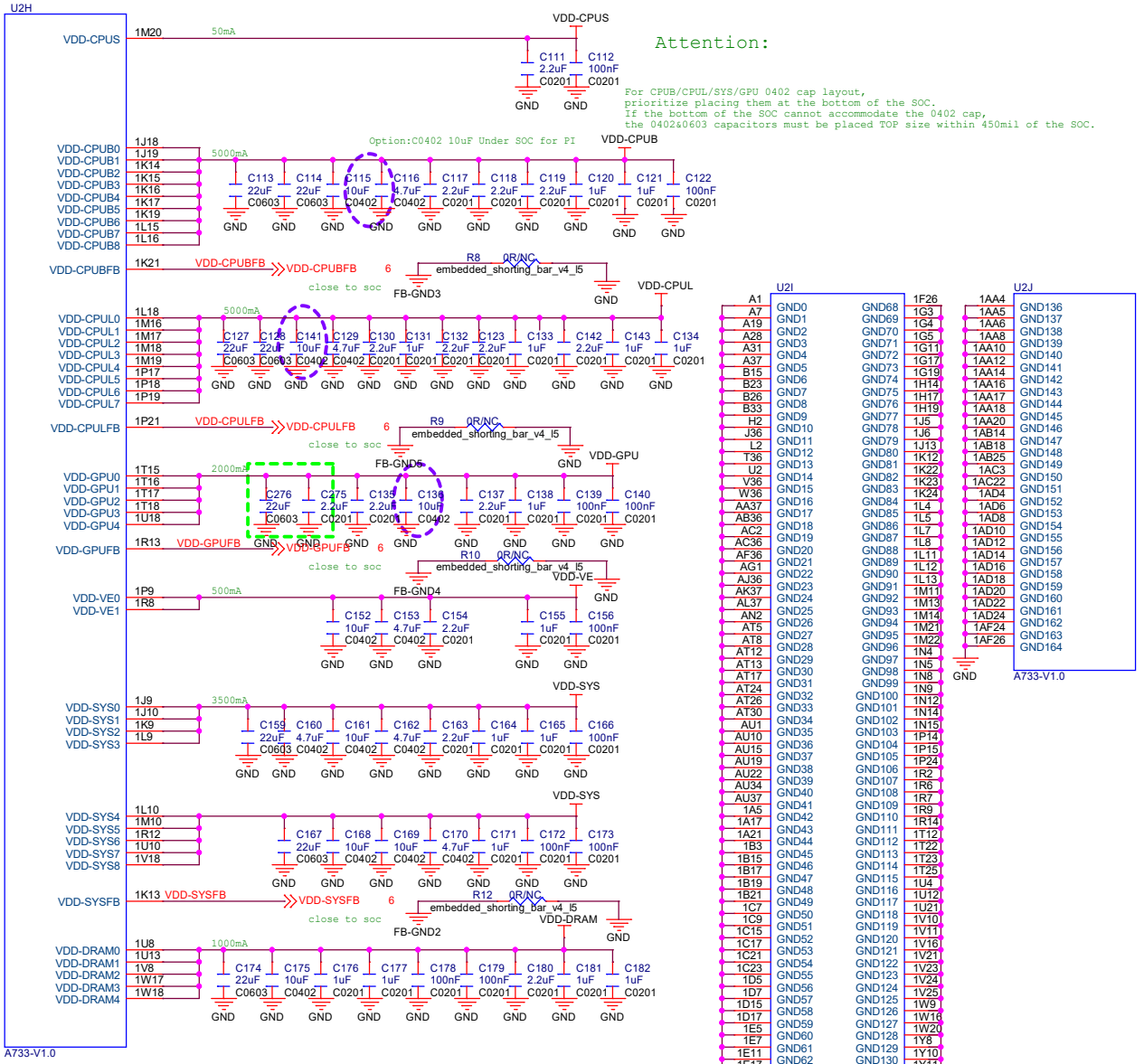
SYS-TEST

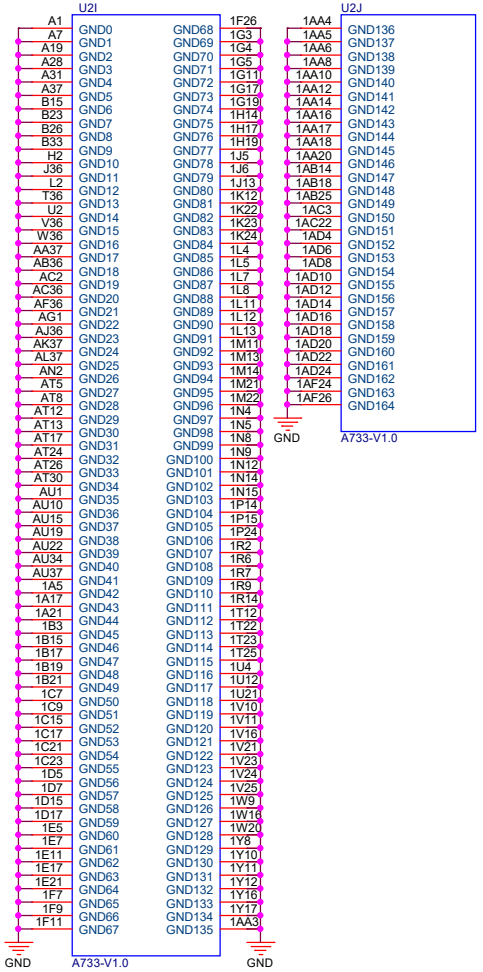
JTAT-SEL
1:software choose JTAG from PF(Default) or PB
0:Force JTAG from PB



default 26M	SEL1	SEL0
SOC=26M	1	0

CORE POWER





Shenzhen Xunlong Software Co., Ltd	
Schematic Name:	OPi ZERO 3W
Page Name:	08_A733 SYS/POWER
Page Size: A3	Sheet: 8 Of: 18 Rev: V1.2

GPIO

U2B

VCC-IO
R16 4.7K 5% R0201
R17 4.7K 5% R0201
9,18 EXT_TWI0_SDA
9,18 EXT_TWI0_SCK

PB: 3.3V VCC-IO供电

18 PB0 PB0 AE1
18 PB1 PB1 AE2
9,18 EXT_TWI0_SCK PB2 1U2
9,18 EXT_TWI0_SDA PB3 1U3
PB4 PWM0-0 AF2
18 I2S0-MCLK PB5 AG2
18 I2S0-BCLK PB6 AH1
18 I2S0-LRCLK PB7 AH2
18 I2S0-DOUT0 PB8 AJ2
18 I2S0-DIN0 1W4
CPUX DEBUG 18 CPUX-TX 1W3
18 CPUX-RX

PB0/UART2-TX/UART0-TX/SPI2-CS0/DSI-TRIG-LCD-TE1/LCD0-D0/JTAG-MS/PB-EINT0
PB1/UART2-RX/UART0-RX/SPI2-CLK/LCD0-D1/JTAG-CK/PB-EINT1
PB2/UART2-RTS/SPI2-MOSI/HDMI-SCL/LCD0-D8/JTAG-DO/TWI0-SCK/PB-EINT2
PB3/UART2-CTS/SPI2-MISO/HDMI-SDA/LCD0-D9/JTAG-DI/TWI0-SDA/PB-EINT3
PB4/PWM0-0/I2S0-MCLK/SPI2-CS1/HDMI-CEC/LCD0-D16/TRACE-CLK/TWI1-SCK/PB-EINT4
PB5/I2S0-BCLK/SPI2-CS2/PWM0-1/LCD0-D17/TRACE-DATA0/TWI1-SDA/PB-EINT5
PB6/CLK-FANOUT1/I2S0-LRCK/SPI2-CS3/PWM0-2/PWM0-8/TRACE-DATA1/PB-EINT6
PB7/CLK-FANOUT2/I2S0-DOUT0/I2S0-DIN1/PWM0-9/OWA0-IN/TRACE-DATA2/TWI1-SCK/PB-EINT7
PB8/CLK-FANOUT3/I2S0-DIN0/I2S0-DOUT1/PWM1-0/OWA0-OUT/TRACE-DATA3/TWI1-SDA/PB-EINT8
PB9/UART0-TX/I2S0-DIN2/I2S0-DOUT2/PWM1-1/WATCHDOG-SIG/LCD0-D16/TWI8-SCK/TWI0-SCK/PB-EINT9
PB10/UART0-RX/I2S0-DIN3/I2S0-DOUT3/PWM1-2/PLL-LOCK-DBG/LCD0-D17/TWI8-SDA/TWI0-SDA/PB-EINT10

3 eMMC-DS eMMC-DS 1AD26
3 eMMC-RST eMMC-RST 1AD25
1AB23
1AB24
1Y22
OK 3 eMMC-CLK R21 33R R0201 AK36
3 eMMC-CMD close AL36
1Y23
3 eMMC-D3 AT37
3 eMMC-D4 AT36
3 eMMC-D0 AR36
3 eMMC-D5 AP36
1AB26
3 eMMC-D1 AP37
3 eMMC-D6 AN37
3 eMMC-D2 AN36
3 eMMC-D7 AM36

PC0/NAND-WE/SDC2-DS/SDC3-DS/PC-EINT0
PC1/NAND-ALE/SDC2-RST/SDC3-RST/PC-EINT1
PC2/NAND-CLE/SPI0-MOSI/SPIF0-MOSI/PC-EINT2
PC3/NAND-CE1/SPI0-CS0/SPIF0-CS0/PC-EINT3
PC4/NAND-CE0/SPI0-MISO/SPIF0-MISO/PC-EINT4
PC5/NAND-RE/SDC2-CLK/SDC3-CLK/PC-EINT5
PC6/NAND-RB0/SDC2-CMD/SDC3-CMD/PC-EINT6
PC7/NAND-RB1/SPI0-CS1/SPIF0-DQS/PC-EINT7
PC8/NAND-DQ7/SDC2-D3/SDC3-D3/SPIF0-D7/PC-EINT8
PC9/NAND-DQ6/SDC2-D4/SDC3-D4/SPIF0-D6/PC-EINT9
PC10/NAND-DQS/SDC2-D2/SDC3-D0/SPIF0-D5/PC-EINT10
PC11/NAND-DQ4/SDC2-D5/SDC3-D5/SPIF0-D4/PC-EINT11
PC12/NAND-DQS/SPI0-CLK/SPIF0-CLK/PC-EINT12
PC13/NAND-DQ3/SDC2-D1/SDC3-D1/PC-EINT13
PC14/NAND-DQ2/SDC2-D6/SDC3-D6/PC-EINT14
PC15/NAND-DQ1/SDC2-D2/SDC3-D2/SPI0-WP/SPIF0-WP/PC-EINT15
PC16/NAND-DQ0/SDC2-D7/SDC3-D7/SPI0-HOLD/SPIF0-HOLD/PC-EINT16

1.8V VCC-PC 60mA 1Y21
VCC-PC

18 PD0 PWM0-0 D2
18 PD1 PWM0-1 F2
18 PD2 PWM0-2 F1
18 PD3 PWM0-3 G1
18 PD4 PWM0-4 G2
5 FAN_EN_PWM PWM0-5 J1
J2
K1
K2
J4

PD0/LCD0-D2/LVDS0-D0P/DSI0-D0P/EINK-D0/PWM0-0/PD-EINT0
PD1/LCD0-D3/LVDS0-D0N/DSI0-D0N/EINK-D1/PWM0-1/PD-EINT1
PD2/LCD0-D4/LVDS0-D1P/DSI0-D1P/EINK-D2/PWM0-2/PD-EINT2
PD3/LCD0-D5/LVDS0-D1N/DSI0-D1N/EINK-D3/PWM0-3/PD-EINT3
PD4/LCD0-D6/LVDS0-D2P/DSI0-CKP/EINK-D4/PWM0-4/PD-EINT4
PD5/LCD0-D7/LVDS0-D2N/DSI0-CKN/EINK-D5/PWM0-5/PD-EINT5
PD6/LCD0-D10/LVDS0-CKP/DSI0-D2P/EINK-D6/PWM0-6/PD-EINT6
PD7/LCD0-D11/LVDS0-CKN/DSI0-D2N/EINK-D7/PWM0-7/PD-EINT7
PD8/LCD0-D12/LVDS0-D3P/DSI0-D3P/EINK-D8/PWM0-8/PD-EINT8
PD9/LCD0-D13/LVDS0-D3N/DSI0-D3N/EINK-D9/PWM0-9/PD-EINT9

17 DSI1-D0P L3
17 DSI1-D0N 1G2
17 DSI1-D1P 1G1
17 DSI1-D1N 1J1
OK 17 DSI1-CKP 1J2
17 DSI1-CKN M2
17 DSI1-D2P M1
17 DSI1-D2N N2
17 DSI1-D3P N1
17 DSI1-D3N L1
16 PCIE-CLKREQn L2
16 PCIE-WAKEn 1L1
16 PCIE-PERSTn P2
16 PCIE_PWREN_H P1

PD10/LCD0-D14/LVDS1-D0P/DSI1-D0P/EINK-D10/SPI1-CS0/DBI-CSX/PWM1-0/PD-EINT10
PD11/LCD0-D15/LVDS1-D0N/DSI1-D0N/EINK-D11/SPI1-CLK/DBI-SCLK/PWM1-1/PD-EINT11
PD12/LCD0-D18/LVDS1-D1P/DSI1-D1P/EINK-D12/SPI1-MOSI/DBI-SDIO/PWM1-2/PD-EINT12
PD13/LCD0-D19/LVDS1-D1N/DSI1-D1N/EINK-D13/SPI1-MISO/DBI-SDI/DBI-TE/DBI-DCX/PWM1-3/PD-EINT13
PD14/LCD0-D20/LVDS1-D2P/DSI1-CKP/EINK-D14/SPI1-HOLD/DBI-DCX/DBI-WRX/-UART3-RTS/PD-EINT14
PD15/LCD0-D21/LVDS1-D2N/DSI1-CKN/EINK-D15/SPI1-CS1/UART3-CTS/PD-EINT15
PD16/LCD0-D22/LVDS1-CKP/DSI1-D2P/EINK-OEH/TWI2-SCK/UART3-TX/PD-EINT16
PD17/LCD0-D23/LVDS1-CKN/DSI1-D2N/EINK-LEH/TWI2-SDA/UART3-RX/PD-EINT17
PD18/LCD0-CLK/LVDS1-D3P/DSI1-D3P/EINK-CKH/SPI1-WP/-DBI-TE-/-UART4-RTS/PD-EINT18
PD19/LCD0-DE/LVDS1-D3N/DSI1-D3N/PCIE-8THPWM1-5/UART4-CTS/PD-EINT19
PD20/LCD0-HSYNC/DSI-TRIG-LCD-TE1/TWI0-SCK/EINK-CKV /PCIE-CLKREQN/UART4-TX/TWI3-SCK/PWM0-2/PD-EINT20
PD21/LCD0-VSYNC/TWI0-SDA/EINK-MODE/PCIE-WAKEN/UART4-RX/TWI3-SDA/PWM0-3/PD-EINT21
PD22/PWM0-4/EINK-SDDO/PCIE-PERSTN/TWI0-SCK/TWI2-SCK/PWM1-4/PD-EINT22
PD23/PWM0-5/PCIE-CLKREQN/TWI0-SDA/TWI2-SDA/PWM1-5/PD-EINT23

3.3V VCC-PD 100mA 1N7
1.8V VCC-LVDS0 60mA 1N6
VCC-PD
VCC-LVDS

18 SPI3_CS0 PE0 A6
18 SPI3_CLK PE1 B6
18 SPI3_MOSI PE2 B5
18 SPI3_MISO PE3 B4
18 SPI3_CS1 PE4 A4
PE5 1B5
13 PE5 PE6 1C5
13 PE6 PE7 B3
CC、CAM 9,13 TWI11_SCK PE8 B2
9,13 TWI11_SDA PE9 1A3
13 PE9 PE10 1A1
13 PE10 PE11 B1
18 PE11 PE12 1C3
18 UART1-TX PE13 C2
18 UART1-RX PE14 C1
9,13 TWI9_SCK PE15 D1
9,13 TWI9_SDA
3.3V VCC-PE 60mA 1G7

PE0/MCSI0-MCLK/CSI0-XHS/SPI3-CS0/LPC-LAD0/NCSI0-HSYNC/UART6-TX/PE-EINT0
PE1/TWI2-SCK/CSI1-XHS/SPI3-CLK/LPC-LPME/NCSI0-VSYNC/UART6-RX/PE-EINT1
PE2/TWI2-SDA/CSI0-XVS-FSYNC/SPI3-MOSI/LPC-LPCPD/NCSI0-PCLK/UART6-RTS/PE-EINT2
PE3/TWI3-SCK/CSI1-XVS-FSYNC/PWM0-0/SPI3-MISO/LPC-LFRAME/NCSI0-D1/UART6-CTS/PE-EINT3
PE4/TWI3-SDA/LEDC/PWM0-1/SPI3-CS1/LPC-LSMI/NCSI0-D0/UART6-DCD/PE-EINT4
PE5/MCSI1-MCLK/PLL-LOCK-DBG/PWM0-2/LPC-LCLK/NCSI0-MCLK/UART6-DSR/PE-EINT5
PE6/CLK-FANOUT1/HDMI-CEC/I2S3-DIN0/I2S5-DOUT1/LPC-SERIRQ/NCSI0-D2/UART6-DTR/PE-EINT6
PE7/CLK-FANOUT2/HDMI-SCL/I2S3-BCLK/LPC-LAD1/NCSI0-D3/TWI11-SCK/PE-EINT7
PE8/CLK-FANOUT3/HDMI-SDA/I2S3-LRCK/LPC-LAD2/NCSI0-D4/TWI11-SDA/PE-EINT8
PE9/MCSI2-MCLK/TCON-FSYNC0/I2S3-MCLK/LPC-LRESET/NCSI0-D5/UART6-RI/PE-EINT9
PE10/TWI4-SCK/TCON-FSYNC1/I2S3-DIN3/I2S3-DOUT3/LPC-LAD3/NCSI0-D6/UART1-RTS/PE-EINT10
PE11/TWI4-SDA/SPI3-MISO/I2S3-DIN2/I2S3-DOUT2/PCIE-CLKREQN/NCSI0-D7/UART1-CTS/PE-EINT11
PE12/SPI3-CS0/I2S3-DOUT0/I2S3-DIN1/PCIE-WAKEN/UART1-TX/PE-EINT12
PE13/SPI3-CLK/PCIE-PERSTN/UART1-RX/PE-EINT13
PE14/SPI3-MOSI/PWM1-8/PCIE-CLKREQN/IR-RX/TWI9-SCK/PE-EINT14
PE15/SPI3-CS1/PWM1-9/IR-RX/TWI9-SDA/PE-EINT15

VCC-PE

12 SDC0-D1 1U1
12 SDC0-CMD AD2
OK 12 SDC0-CLK R22 33R R0201 AD1
12 SDC0-D0 AB2
12 SDC0-D3 1R1
12 SDC0-D2 1R4
12 SDC0-DET 1R3
3.3V VCC-IO 150mA 1U5
VCC18-PF 60mA 1U6

PF0/SDC0-D1/JTAG-MS/TRACE-DATA3/UART5-RTS/TWI2-SCK/PF-EINT0
PF1/SDC0-CMD/JTAG-DO/TRACE-DATA2/PF-EINT1
PF2/SDC0-CLK/UART0-TX/TRACE-CLK/PF-EINT2
PF3/SDC0-D0/JTAG-DI/TRACE-DATA0/UART5-CTS/TWI2-SDA/PF-EINT3
PF4/SDC0-D3/UART0-RX/UART5-RX/PF-EINT4
PF5/SDC0-D2/JTAG-CK/TRACE-DATA1/UART5-TX/PF-EINT5
PF6/PF-EINT6

VCC-IO
VCC18-PF

A733-V1.0

PG0/SDC1-CLK/LCD0-D0/PWM1-1/LPC-LAD0/PG-EINT0
PG1/SDC1-CMD/LCD0-D1/PWM1-2/LPC-LPME/PG-EINT1
PG2/SDC1-D0/LCD0-D8/PWM1-3/LPC-LPCPD/PG-EINT2
PG3/SDC1-D1/LCD0-D9/PWM1-4/LPC-LFRAME/PG-EINT3
PG4/SDC1-D2/LCD0-D16/PWM1-5/LPC-LSMI/PG-EINT4
PG5/SDC1-D3/LCD0-D17/PWM1-6/LPC-LCLK/PG-EINT5
PG6/UART1-TX/UART4-TX/PWM1-7/LPC-SERIRQ/PG-EINT6
PG7/UART1-RX/UART4-RX/PWM1-8/LPC-LAD1/PG-EINT7
PG8/UART1-RTS/UART4-RTS/UART5-TX/LPC-LAD2/TWI0-SCK/PG-EINT8
PG9/UART1-CTS/UART4-CTS/UART5-RX/LPC-LRESET/TWI0-SDA/PG-EINT9
PG10/I2S1-MCLK/CLK-FANOUT0/PWM1-9/LPC-LAD3/DMIC-CLK/PG-EINT10
PG11/I2S1-BCLK/TWI12-SCK/TWI9-SCK/SGPIO-LOAD/DMIC-DATA3/HDMI-CEC/PG-EINT11
PG12/I2S1-LRCK/TWI12-SDA/TWI9-SDA/SGPIO-CLK/DMIC-DATA2/HDMI-SCL/PG-EINT12
PG13/I2S1-DOUT0/I2S1-DIN1/TWI9-SCK/SGPIO-DATAIN/DMIC-DATA1/HDMI-SDA/PG-EINT13
PG14/I2S1-DIN0/I2S1-DOUT1/TWI6-SDA/SGPIO-DATAOUT/DMIC-DATA0/PG-EINT14

VCC-PG

PH0/TWI0-SCK/I2S2-DOUT2/I2S2-DIN2/RGMII0-RXD1/RMII0-RXD1/TWI3-SCK/SPI1-HOLD/PH-EINT0
PH1/TWI0-SDA/I2S2-DOUT3/I2S2-DIN3/RGMII0-RXD0/RMII0-RXD0/TWI3-SDA/SPI1-WP/PH-EINT1
PH2/TWI1-SCK/I2S2-MCLK/RGMII0-RXCTL/RMII0-CRS-DV/SPI1-CS1/PH-EINT2
PH3/TWI1-SDA/I2S2-BCLK/RGMII0-CLKIN/RMII0-RXER/SPI1-MISO/PH-EINT3
PH4/UART3-TX/SPI2-CS3/I2S2-LRCK/RGMII0-TXD1/RMII0-TXD1/SPI1-CS0/PH-EINT4
PH5/UART3-RX/I2S2-DIN1/I2S2-DOUT0/RGMII0-TXD0/RMII0-TXD0/SPI1-MOSI/PH-EINT5
PH6/UART3-RTS/I2S2-DOUT1/I2S2-DIN0/RGMII0-TXCK/RMII0-TXCK/SPI1-CLK/PH-EINT6
PH7/UART3-CTS/SPI2-CS2/DSI-TRIG-LCD-TE1/RGMII0-TXCTL/RMII0-TXEN/PH-EINT7
PH8/DMIC-CLK/SPI2-CLK/RGMII0-MDC/PH-EINT8
PH9/DMIC-DATA0/SPI2-CS0/RGMII0-MDIO/PH-EINT9
PH10/DMIC-DATA1/SPI2-MOSI/RGMII0-RXD3/RMII0-NULL/PH-EINT10
PH11/DMIC-DATA2/SPI2-MISO/RGMII0-RXD2/RMII0-NULL/PH-EINT11
PH12/DMIC-DATA3/SPI2-CS1/RGMII0-RXCK/RMII0-NULL/PH-EINT12
PH13/UART5-TX/RGMII0-TXD3/RMII0-NULL/TWI7-SDA/PH-EINT13
PH14/UART5-RX/RGMII0-TXD2/RMII0-NULL/TWI7-SCK/PH-EINT14
PH15/UART5-RTS/LEDC/DSI-TRIG-LCD-TE1/RGMII0-EPHY-25/50M/PH-EINT15
PH16/UART5-CTS/IR-TX/PH-EINT16

VCC-PH

PJ22/PWM1-4/UART3-TX/UART2-RTS/TWI7-SCK/TWI3-SCK/TWI11-SCK/LCD0-D8/PJ-EINT22
PJ23/PWM1-5/UART3-RX/UART2-CTS/TWI7-SDA/TWI3-SDA/TWI11-SDA/LCD0-D9/PJ-EINT23
PJ24/PWM1-6/UART4-TX/TWI4-SCK/SPI3-CLK/PJ-EINT24
PJ25/PWM1-7/UART4-RX/TWI4-SDA/SPI3-MOSI/PJ-EINT25
PJ26/PWM1-8/UART4-RTS/UART2-TX/TWI5-SCK/SPI3-MISO/LCD0-D16/PJ-EINT26
PJ27/PWM1-9/UART4-CTS/UART2-RX/DSI-TRIG-LCD-TE1/TWI5-SDA/SPI3-CS0/LCD0-D17/PJ-EINT27

VCC-PJ

PK0/MCSIA-D0N/UART6-DCD/I2S4-BCLK/HDMI-CEC/TWI11-SDA/NCSI1-HSYNC/SGPIO-SLOAD/PK-EINT0
PK1/MCSIA-D0P/UART6-DSR/I2S4-MCLK/HDMI-SCL/TWI11-SCK/NCSI1-VSYNC/SGPIO-SCLK/PK-EINT1
PK2/MCSIA-D1N/UART6-DTR/I2S4-LRCK/HDMI-SDA/TWI5-SDA/SDA/NCSI1-PCLK/SGPIO-SDATAIN/PK-EINT2
PK3/MCSIA-D1P/UART6-RI/I2S4-DIN0/I2S4-DOUT1/TWI5-SCK/NCSI1-MCLK/SGPIO-SDATAOUT/PK-EINT3
PK4/MCSIA-CKN/PCIE-WAKEN/I2S4-DOUT0/I2S4-DIN1/SPI3-CS0/NCSI1-D15/JTAG-MAS-MS/PK-EINT4
PK5/MCSIA-CKP/PCIE-CLKREQN/PWM1-8/PWM1-9/SPI3-CLK/NCSI1-D14/JTAG-MAS-CK/PK-EINT5
PK6/MCSIA-D2N/TWI2-SCK/UART4-TX/UART2-RTS/SPI3-MOSI/NCSI1-D13/JTAG-MAS-DO/PK-EINT6
PK7/MCSIA-D2P/TWI2-SDA/UART4-RX/UART2-CTS/SPI3-MISO/NCSI1-D12/JTAG-MAS-DI/PK-EINT7
PK8/MCSIA-D3N/MCSI0-MCLK/UART4-RTS/UART2-TX/SPI3-CS1/NCSI1-D11/JTAG-MAS-NTRST/PK-EINT8
PK9/MCSIA-D3P/MCSI2-MCLK/UART4-CTS/UART2-RX/NCSI1-D10/JTAG-MAS-SRST/PK-EINT9
PK10/MCSIB-D0N/UART6-TX/PWM0-3/NCSI1-D9/PK-EINT10
PK11/MCSIB-D0P/UART6-RX/PWM0-4/NCSI1-D8/PK-EINT11
PK12/MCSIB-D1N/UART6-RTS/PWM0-5/NCSI1-D7/PK-EINT12
PK13/MCSIB-D1P/UART6-CTS/PWM0-6/NCSI1-D6/PK-EINT13
PK14/MCSIB-CKN/PCIE-PERSTN/PWM0-7/NCSI1-D5/PK-EINT14
PK15/MCSIB-CKP/PWM0-8/NCSI1-D4/PK-EINT15
PK16/MCSIB-D2N/TWI3-SCK/UART2-TX/PWM0-9/NCSI1-D3/PK-EINT16
PK17/MCSIB-D2P/TWI3-SDA/UART2-RX/PWM1-0/NCSI1-D2/PK-EINT17
PK18/MCSIB-D3N/MCSI1-MCLK/UART2-RTS/PWM1-1/NCSI1-D1/TWI9-SCK/PK-EINT18
PK19/MCSIB-D3P/MCSI2-MCLK/UART2-CTS/PWM1-2/NCSI1-D0/TWI9-SDA/PK-EINT19
PK20/MCSIC-D0N/TWI2-SCK/UART3-TX/UART1-RTS/PWM1-3/NCSI1-PCLK/UART1-RTS/PK-EINT20
PK21/MCSIC-D0P/TWI2-SDA/UART3-RX/PWM0-2/NCSI1-MCLK/UART1-CTS/PK-EINT21
PK22/MCSIC-D1N/TWI3-SCK/UART3-RTS/PWM1-6/NCSI1-HSYNC/UART1-TX/PK-EINT22
PK23/MCSIC-D1P/TWI3-SDA/UART3-CTS/PWM1-7/NCSI1-VSYNC/UART1-RX/PK-EINT23
PK24/MCSIC-CKN/MCSI0-MCLK/PWM0-6/TWI12-SCK/TWI10-SDA/PK-EINT24
PK25/MCSIC-CKP/MCSI1-MCLK/PWM0-7/TWI12-SDA/TWI10-SCK/PK-EINT25

VCC-PK
VCC-MCSI

PL0/S-TWI0-SCK/S-TWI1-SCK/S-TWI2-SCK/PL-EINT0
PL1/S-TWI0-SDA/S-TWI1-SDA/S-TWI2-SDA/S-IR-RX/PL-EINT1
PL2/S-UART1-TX/S-UART0-TX/S-TWI1-SDA/S-PWM0-0/PL-EINT2
PL3/S-UART1-RX/S-UART0-RX/S-TWI1-SCK/S-IR-RX/S-PWM0-1/PL-EINT3
PL4/S-JTAG-MS/S-TWI2-SCK/S-SPI0-CS0/S-IR-RX/S-PWM0-2/PL-EINT4
PL5/S-JTAG-CK/S-TWI2-SDA/S-SPI0-CLK/S-PWM0-3/PL-EINT5
PL6/S-JTAG-DO/S-UART0-TX/S-SPI0-MOSI/S-IR-RX/S-PWM0-4/PL-EINT6
PL7/S-JTAG-DI/S-UART0-RX/S-SPI0-MISO/S-PWM0-5/PL-EINT7
PL8/S-TWI1-SCK/S-UART1-TX/S-TWI0-SCK/S-TWI2-SCK/S-PWM0-6/PL-EINT8
PL9/S-TWI1-SDA/S-UART1-RX/S-TWI0-SDA/S-TWI2-SDA/S-PWM0-7/PL-EINT9
PL10/S-UART0-TX/S-TWI2-SCK/S-UART1-TX/S-PWM0-8/PL-EINT10
PL11/S-UART0-RX/S-TWI2-SDA/S-UART1-RX/S-IR-RX/S-PWM0-9/PL-EINT11
PL12/S-UART1-TX/S-TWI1-SCK/S-TWI2-SCK/S-IR-RX/PL-EINT12
PL13/S-UART1-RX/S-TWI1-SDA/S-TWI2-SDA/PL-EINT13

VCC-PL

PM0/S-JTAG-MS/S-SPI0-CS0/S-RJTAG-MS/S-TWI1-SCK/S-PWM0-2/S-IR-RX/PM-EINT0
PM1/S-JTAG-CK/S-SPI0-CLK/S-RJTAG-CK/S-TWI1-SDA/S-PWM0-3/PM-EINT1
PM2/S-JTAG-DO/S-SPI0-MOSI/S-UART0-TX/S-UART1-TX/S-PWM0-4/PM-EINT2
PM3/S-JTAG-DI/S-SPI0-MISO/S-UART0-RX/S-UART1-RX/S-PWM0-5/PM-EINT3
PM4/S-UART0-TX/S-TWI2-SCK/S-TWI1-SCK/S-UART1-TX/S-PWM0-0/S-IR-RX/PM-EINT4
PM5/S-UART0-RX/S-TWI2-SDA/S-TWI1-SDA/S-UART1-RX/S-PWM0-1/S-IR-RX/PM-EINT5

VCC-PM

1N3 WL-SDIO-CLK 14
1N2 WL-SDIO-CMD 14
1N1 WL-SDIO-D0 14
R2 WL-SDIO-D1 14
R1 WL-SDIO-D2 14
T2 WL-SDIO-D3 14
T1 14
PG6/UART1-TX/UART4-TX/PWM1-7/LPC-SERIRQ/PG-EINT6 V2 BT-UART-RX 14
PG7/UART1-RX/UART4-RX/PWM1-8/LPC-LAD1/PG-EINT7 V1 BT-UART-TX 14
PG8/UART1-RTS/UART4-RTS/UART5-TX/LPC-LAD2/TWI0-SCK/PG-EINT8 W2 BT-UART-CTS 14
PG9/UART1-CTS/UART4-CTS/UART5-RX/LPC-LRESET/TWI0-SDA/PG-EINT9 W1 BT-UART-RTS 14
PG10/I2S1-MCLK/CLK-FANOUT0/PWM1-9/LPC-LAD3/DMIC-CLK/PG-EINT10 Y2
PG11/I2S1-BCLK/TWI12-SCK/TWI9-SCK/SGPIO-LOAD/DMIC-DATA3/HDMI-CEC/PG-EINT11 AA2 I2S1_BCLK 14
PG12/I2S1-LRCK/TWI12-SDA/TWI9-SDA/SGPIO-CLK/DMIC-DATA2/HDMI-SCL/PG-EINT12 AB1 I2S1_LRCK 14
PG13/I2S1-DOUT0/I2S1-DIN1/TWI9-SCK/SGPIO-DATAIN/DMIC-DATA1/HDMI-SDA/PG-EINT13 AA1 I2S1-DOUT0 14 OK
PG14/I2S1-DIN0/I2S1-DOUT1/TWI6-SDA/SGPIO-DATAOUT/DMIC-DATA0/PG-EINT14 I2S1_DIN0 14
1R5 70mA VCC-PG 1.8V

PH0/TWI0-SCK/I2S2-DOUT2/I2S2-DIN2/RGMII0-RXD1/RMII0-RXD1/TWI3-SCK/SPI1-HOLD/PH-EINT0 AK1
PH1/TWI0-SDA/I2S2-DOUT3/I2S2-DIN3/RGMII0-RXD0/RMII0-RXD0/TWI3-SDA/SPI1-WP/PH-EINT1 AK2
PH2/TWI1-SCK/I2S2-MCLK/RGMII0-RXCTL/RMII0-CRS-DV/SPI1-CS1/PH-EINT2 1AA1
PH3/TWI1-SDA/I2S2-BCLK/RGMII0-CLKIN/RMII0-RXER/SPI1-MISO/PH-EINT3 AL1
PH4/UART3-TX/SPI2-CS3/I2S2-LRCK/RGMII0-TXD1/RMII0-TXD1/SPI1-CS0/PH-EINT4 AM2
PH5/UART3-RX/I2S2-DIN1/I2S2-DOUT0/RGMII0-TXD0/RMII0-TXD0/SPI1-MOSI/PH-EINT5 AN1
PH6/UART3-RTS/I2S2-DOUT1/I2S2-DIN0/RGMII0-TXCK/RMII0-TXCK/SPI1-CLK/PH-EINT6 AL2
PH7/UART3-CTS/SPI2-CS2/DSI-TRIG-LCD-TE1/RGMII0-TXCTL/RMII0-TXEN/PH-EINT7 AP1
PH8/DMIC-CLK/SPI2-CLK/RGMII0-MDC/PH-EINT8 1AC1
PH9/DMIC-DATA0/SPI2-CS0/RGMII0-MDIO/PH-EINT9 1AE1
PH10/DMIC-DATA1/SPI2-MOSI/RGMII0-RXD3/RMII0-NULL/PH-EINT10 1AC2
PH11/DMIC-DATA2/SPI2-MISO/RGMII0-RXD2/RMII0-NULL/PH-EINT11 AT2
PH12/DMIC-DATA3/SPI2-CS1/RGMII0-RXCK/RMII0-NULL/PH-EINT12 AP2
PH13/UART5-TX/RGMII0-TXD3/RMII0-NULL/TWI7-SDA/PH-EINT13 AR2
PH14/UART5-RX/RGMII0-TXD2/RMII0-NULL/TWI7-SCK/PH-EINT14 AT1 PH15
PH15/UART5-RTS/LEDC/DSI-TRIG-LCD-TE1/RGMII0-EPHY-25/50M/PH-EINT15 AU2 PH16 BOARD-ID-SEL-GPIO 10
PH16/UART5-CTS/IR-TX/PH-EINT16 PH16:在 boot 阶段用于参数调制 DDR-PARA-SEL-GPIO 10
1W6 130mA VCC-PH 1.8V

PJ22/PWM1-4/UART3-TX/UART2-RTS/TWI7-SCK/TWI3-SCK/TWI11-SCK/LCD0-D8/PJ-EINT22 1E2 LCD_PWR_EN 17
PJ23/PWM1-5/UART3-RX/UART2-CTS/TWI7-SDA/TWI3-SDA/TWI11-SDA/LCD0-D9/PJ-EINT23 1E1 LCD_RST 17
PJ24/PWM1-6/UART4-TX/TWI4-SCK/SPI3-CLK/PJ-EINT24 1C2
PJ25/PWM1-7/UART4-RX/TWI4-SDA/SPI3-MOSI/PJ-EINT25 1C1 PWM1-7 17
PJ26/PWM1-8/UART4-RTS/UART2-TX/TWI5-SCK/SPI3-MISO/LCD0-D16/PJ-EINT26 1E3
PJ27/PWM1-9/UART4-CTS/UART2-RX/DSI-TRIG-LCD-TE1/TWI5-SDA/SPI3-CS0/LCD0-D17/PJ-EINT27 1E4 LCD_TE 17 1.8V
1L6 60mA VCC-PJ

PK0/MCSIA-D0N/UART6-DCD/I2S4-BCLK/HDMI-CEC/TWI11-SDA/NCSI1-HSYNC/SGPIO-SLOAD/PK-EINT0 1D11 MCSIA-D0N 13
PK1/MCSIA-D0P/UART6-DSR/I2S4-MCLK/HDMI-SCL/TWI11-SCK/NCSI1-VSYNC/SGPIO-SCLK/PK-EINT1 1C11 MCSIA-D0P 13
PK2/MCSIA-D1N/UART6-DTR/I2S4-LRCK/HDMI-SDA/TWI5-SDA/SDA/NCSI1-PCLK/SGPIO-SDATAIN/PK-EINT2 1E11 MCSIA-D1N 13
PK3/MCSIA-D1P/UART6-RI/I2S4-DIN0/I2S4-DOUT1/TWI5-SCK/NCSI1-MCLK/SGPIO-SDATAOUT/PK-EINT3 1A11 MCSIA-D1P 13
PK4/MCSIA-CKN/PCIE-WAKEN/I2S4-DOUT0/I2S4-DIN1/SPI3-CS0/NCSI1-D15/JTAG-MAS-MS/PK-EINT4 1B7 MCSIA-CKN 13
PK5/MCSIA-CKP/PCIE-CLKREQN/PWM1-8/PWM1-9/SPI3-CLK/NCSI1-D14/JTAG-MAS-CK/PK-EINT5 1A7 MCSIA-CKP 13
PK6/MCSIA-D2N/TWI2-SCK/UART4-TX/UART2-RTS/SPI3-MOSI/NCSI1-D13/JTAG-MAS-DO/PK-EINT6 1B9 MCSIA-D2N 13
PK7/MCSIA-D2P/TWI2-SDA/UART4-RX/UART2-CTS/SPI3-MISO/NCSI1-D12/JTAG-MAS-DI/PK-EINT7 1A9 MCSIA-D2P 13
PK8/MCSIA-D3N/MCSI0-MCLK/UART4-RTS/UART2-TX/SPI3-CS1/NCSI1-D11/JTAG-MAS-NTRST/PK-EINT8 1E9 MCSIA-D3N 13
PK9/MCSIA-D3P/MCSI2-MCLK/UART4-CTS/UART2-RX/NCSI1-D10/JTAG-MAS-SRST/PK-EINT9 1D9 MCSIA-D3P 13
PK10/MCSIB-D0N/UART6-TX/PWM0-3/NCSI1-D9/PK-EINT10 B16 MCSIB-D0N 13
PK11/MCSIB-D0P/UART6-RX/PWM0-4/NCSI1-D8/PK-EINT11 B17 MCSIB-D0P 13
PK12/MCSIB-D1N/UART6-RTS/PWM0-5/NCSI1-D7/PK-EINT12 A15 MCSIB-D1N 13 OK
PK13/MCSIB-D1P/UART6-CTS/PWM0-6/NCSI1-D6/PK-EINT13 A16 MCSIB-D1P 13
PK14/MCSIB-CKN/PCIE-PERSTN/PWM0-7/NCSI1-D5/PK-EINT14 B13 MCSIB-CKN 13
PK15/MCSIB-CKP/PWM0-8/NCSI1-D4/PK-EINT15 B14 MCSIB-CKP 13
PK16/MCSIB-D2N/TWI3-SCK/UART2-TX/PWM0-9/NCSI1-D3/PK-EINT16 A12 MCSIB-D2N 13
PK17/MCSIB-D2P/TWI3-SDA/UART2-RX/PWM1-0/NCSI1-D2/PK-EINT17 A13 MCSIB-D2P 13
PK18/MCSIB-D3N/MCSI1-MCLK/UART2-RTS/PWM1-1/NCSI1-D1/TWI9-SCK/PK-EINT18 B11 MCSIB-D3N 13
PK19/MCSIB-D3P/MCSI2-MCLK/UART2-CTS/PWM1-2/NCSI1-D0/TWI9-SDA/PK-EINT19 B12 MCSIB-D3P 13
PK20/MCSIC-D0N/TWI2-SCK/UART3-TX/UART1-RTS/PWM1-3/NCSI1-PCLK/UART1-RTS/PK-EINT20 A10
PK21/MCSIC-D0P/TWI2-SDA/UART3-RX/PWM0-2/NCSI1-MCLK/UART1-CTS/PK-EINT21 B10
PK22/MCSIC-D1N/TWI3-SCK/UART3-RTS/PWM1-6/NCSI1-HSYNC/UART1-TX/PK-EINT22 B9
PK23/MCSIC-D1P/TWI3-SDA/UART3-CTS/PWM1-7/NCSI1-VSYNC/UART1-RX/PK-EINT23 A9 TP_RST 17
PK24/MCSIC-CKN/MCSI0-MCLK/PWM0-6/TWI12-SCK/TWI10-SDA/PK-EINT24 B7 TP_INT 17
PK25/MCSIC-CKP/MCSI1-MCLK/PWM0-7/TWI12-SDA/TWI10-SCK/PK-EINT25 B8 TWI12-SCK 9,17 LCD 9,17 TWI12-SCK
TWI12-SDA 9,17 9,17 TWI12-SDA
1H11 90mA VCC-PK 1.8V
1G9 25mA
VCC-PK
VCC-MCSI

M37 PL0
M36 PL1 PMU-SCK 6
L36 PL2 S-PWM0-0 PMU-SDA 6 PM IC
K36 PL3 S-PWM0-1/IR
K37 PL4 S-TWI2-SCK PL2 18
J37 PL5 S-TWI2-SDA PL3 18
H36 PL4 18
G36 PL5 18
G37 PL6 PWM_LED1 5
F37 PL9 TYPEC_EN 11
1B26 PL10 TYPEC_CC_INT 11
1B25 PL11 SBU1-DC 11
1H26 PL12 SBU2-DC 11
1H25 PL13 S_TWI1_SCK 11
S_TWI1_SDA 11
1M23 50mA VCC-PL 3.3V

T37
R37 WL-WAKE-AP 14
P36 WL-REG-ON 14
R36 AP_WAKE_WL 14
N37 AP-WAKE-BT 14
N36 BT-WAKE-AP 14
1P22 25mA VCC-PM 1.8V

VCC-PK
R1139 2.2K 5% R0201
R1138 2.2K 5% R0201

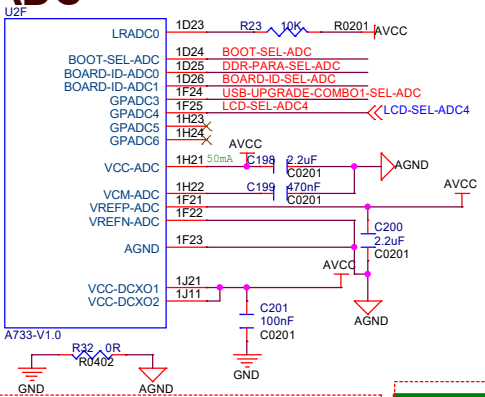
VCC-PE
R18 2.2K 5% R0201
R19 2.2K 5% R0201
9,13 TWI11_SCK
9,13 TWI11_SDA
CAM1

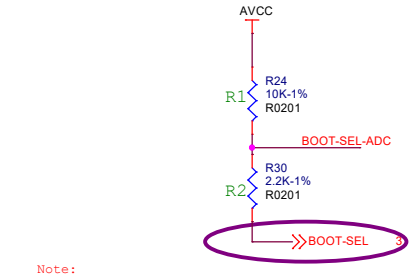
VCC-PE
R1136 2.2K 5% R0201
R1137 2.2K 5% R0201
9,13 TWI9_SCK
9,13 TWI9_SDA
CAM2

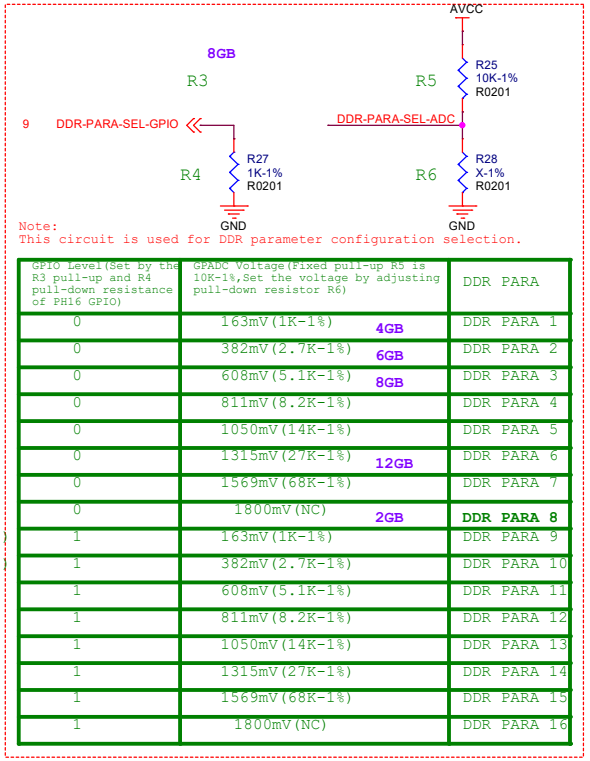
Note PD:
The line sequence of LVDS and MIPI-DSI is inconsistent, as shown in the following table.

GPIO	LVDS Function	MIPI-DSI Function
PD0	LVDS0-D2P	DSI-D2P
PD1	LVDS0-D

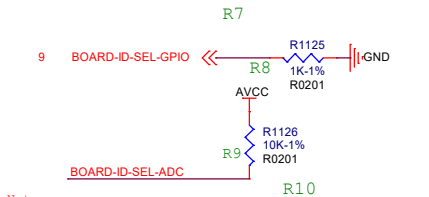
ADC



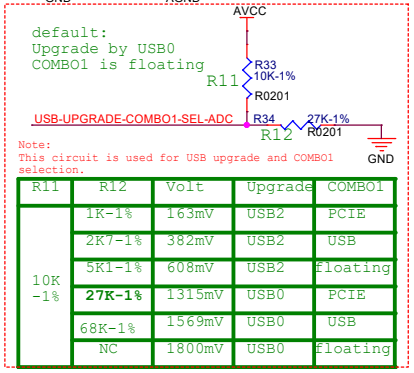




GPIO Level(Set by the R3 pull-up and R4 pull-down resistance of PH16 GPIO)	GPADC Voltage(Fixed pull-up R5 is 10K-1%,Set the voltage by adjusting pull-down resistor R6)		DDR PARA
0	163mV(1K-1%)	4GB	DDR PARA 1
0	382mV(2.7K-1%)	6GB	DDR PARA 2
0	608mV(5.1K-1%)	8GB	DDR PARA 3
0	811mV(8.2K-1%)		DDR PARA 4
0	1050mV(14K-1%)		DDR PARA 5
0	1315mV(27K-1%)	12GB	DDR PARA 6
0	1569mV(68K-1%)		DDR PARA 7
0	1800mV(NC)	2GB	**DDR PARA 8**
1	163mV(1K-1%)		DDR PARA 9
1	382mV(2.7K-1%)		DDR PARA 10
1	608mV(5.1K-1%)		DDR PARA 11
1	811mV(8.2K-1%)		DDR PARA 12
1	1050mV(14K-1%)		DDR PARA 13
1	1315mV(27K-1%)		DDR PARA 14
1	1569mV(68K-1%)		DDR PARA 15
1	1800mV(NC)		DDR PARA 16



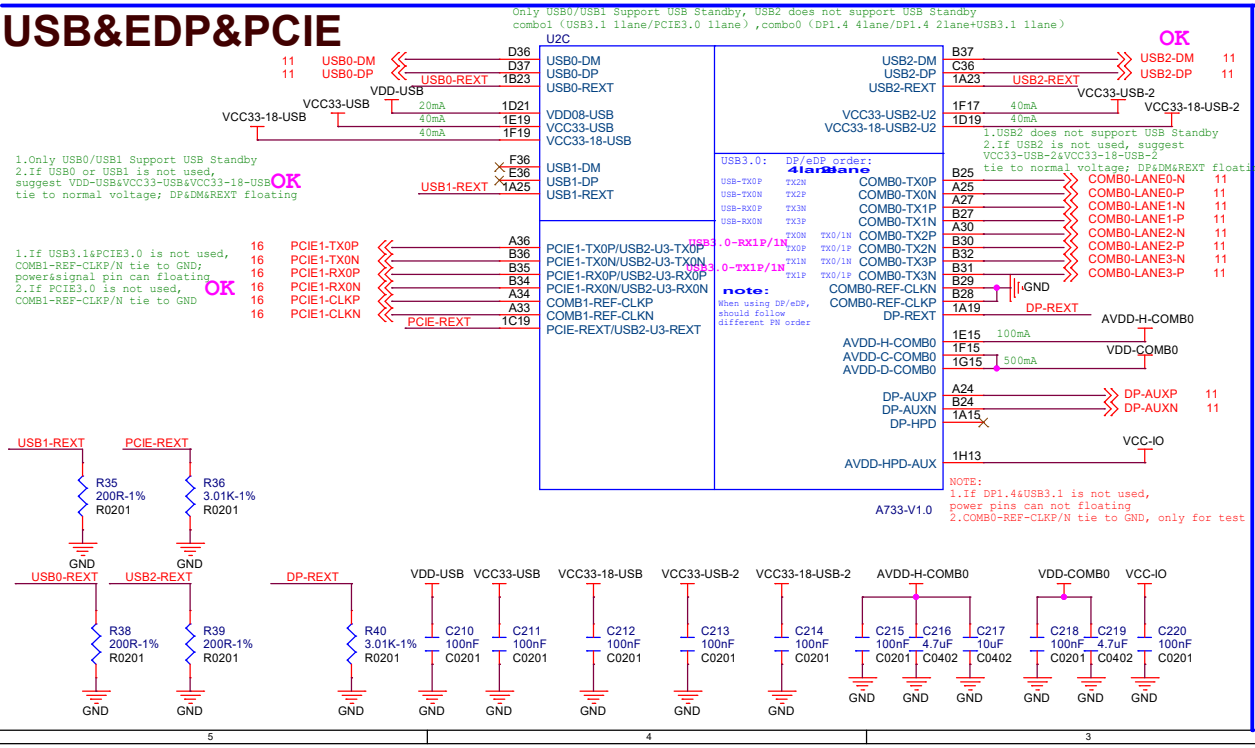
GPIO-SET		BOARD-ID-SEL-ADC			BoardID Config
R7	R8	R9	R10	Volt	
NC	1K pull down	10K -1%	1K-1%	163mV	Config-1
			2K7-1%	382mV	Config-2
			5K1-1%	608mV	Config-3
			8K2-1%	811mV	Config-4
			14K-1%	1050mV	Config-5
			27K-1%	1315mV	Config-6
			68K-1%	1569mV	Config-7
			NC	1800mV	**Config-8**
47K pull up	NC	10K -1%	1K-1%	163mV	Config-9
			2K7-1%	382mV	Config-10
			5K1-1%	608mV	Config-11
			8K2-1%	811mV	Config-12
			14K-1%	1050mV	Config-13
			27K-1%	1315mV	Config-14
			68K-1%	1569mV	Config-15
			NC	1800mV	Config-16



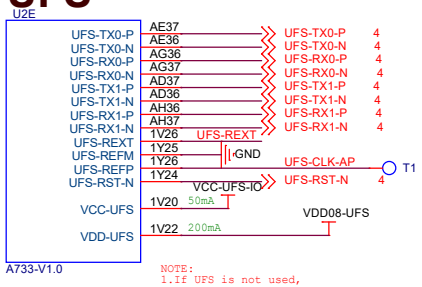
R11	R12	Volt	Upgrade	COMBO1
10K -1%	1K-1%	163mV	USB2	PCIE
	2K7-1%	382mV	USB2	USB
	5K1-1%	608mV	USB2	floating
	27K-1%	1315mV	USB0	PCIE
	68K-1%	1569mV	USB0	USB
	NC	1800mV	USB0	floating

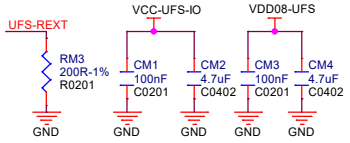
BOOT-SEL-ADC			Boot Media Type Select
R1-1%	R2-1%	Volt	
NC	10K	0mV	SMHC0->MLC NAND->SLC NAND->TRY(except SPI4 in PI)
10K	1K	163mV	SMHC0->SLC NAND->MLC NAND->TRY(except SPI4 in PI)
10K	**2K2**	325mV	SMHC0->EMMC_USER->EMMC_BOOT->TRY(except SPI4 in PI)
10K	3K9	505mV	SMHC0->EMMC_BOOT->EMMC_USER->TRY(except SPI4 in PI)
6K8	4K7	736mV	SMHC0->SPI0 NOR->TRY(except SPI4 in PI)
6K8	6K8	900mV	SMHC0->SPI0 NAND->TRY(except SPI4 in PI)
4K3	6K8	1102mV	SMHC0->SPI4 NOR in PI->TRY(except SPI4 in PI)
2K2	10K	1475mV	SMHC0->SPI4 NAND in PI->TRY(except SPI4 in PI)
10K	NC	1800mV	SMHC0->UFS->TRY(except SPI4 in PI)

USB&EDP&PCIE

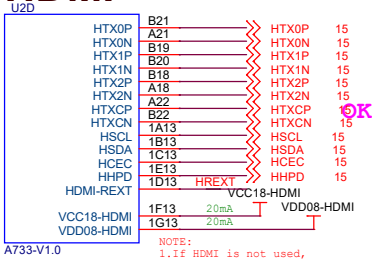


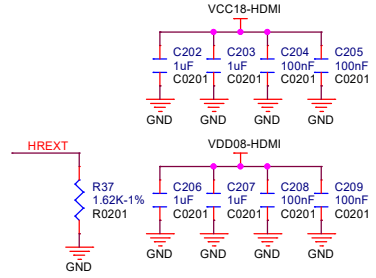
UFS



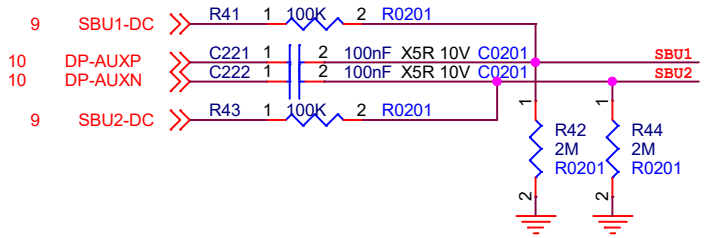


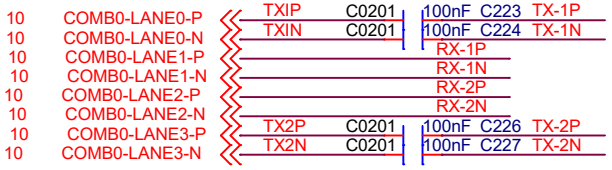
HDMI

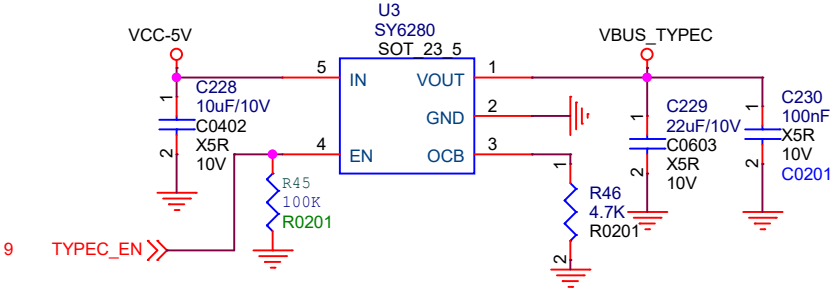




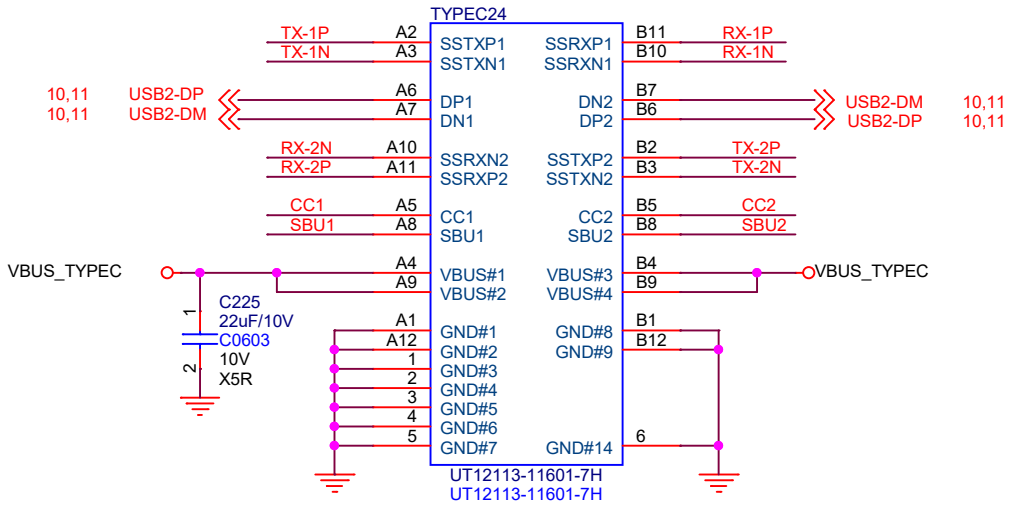
Shenzhen Xunlong Software Co., Ltd			
Schematic Name:	OPi ZERO 3W		
Page Name:	10_A733 Analog&High-speed		
Page Size:	A3	Sheet: 10 Of: 18	Rev: V1.2



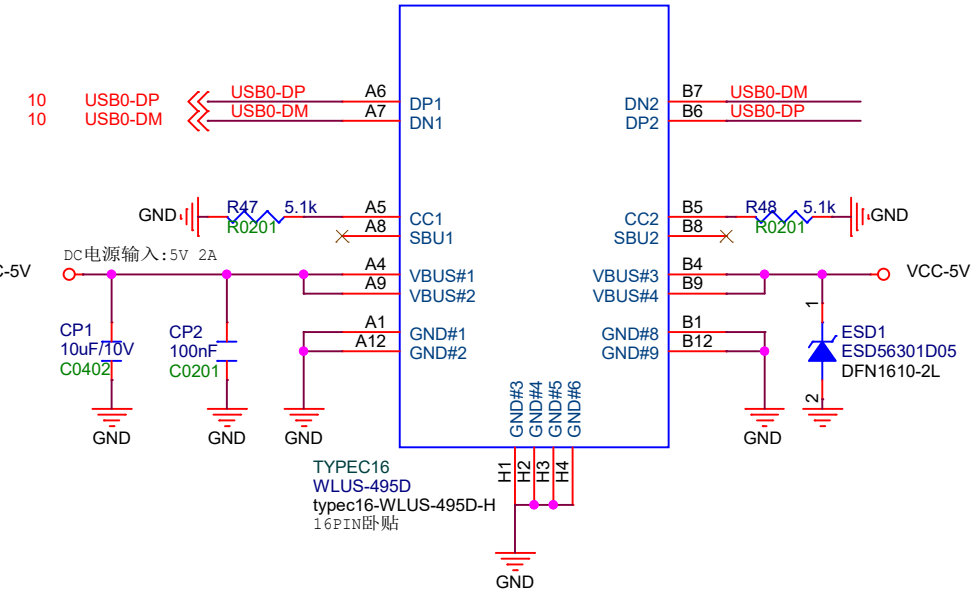




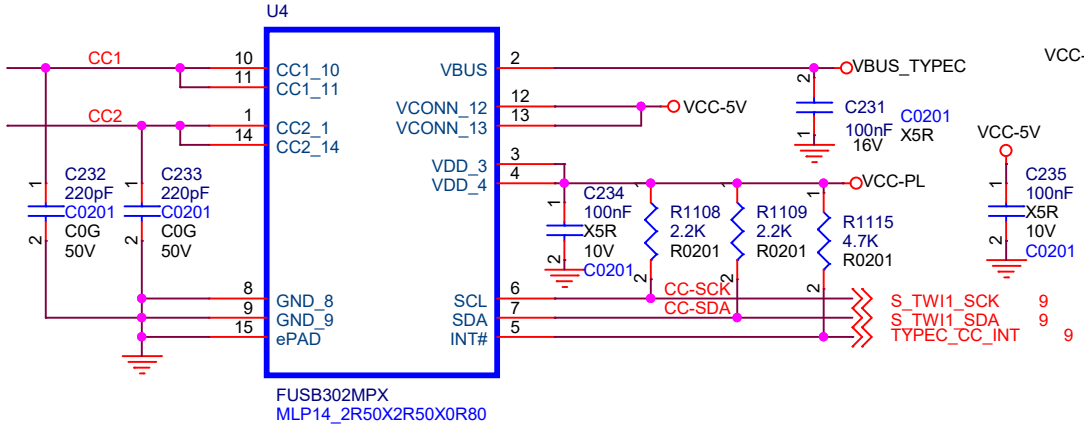
TypeC DP Port



TPYE-C POWER IN

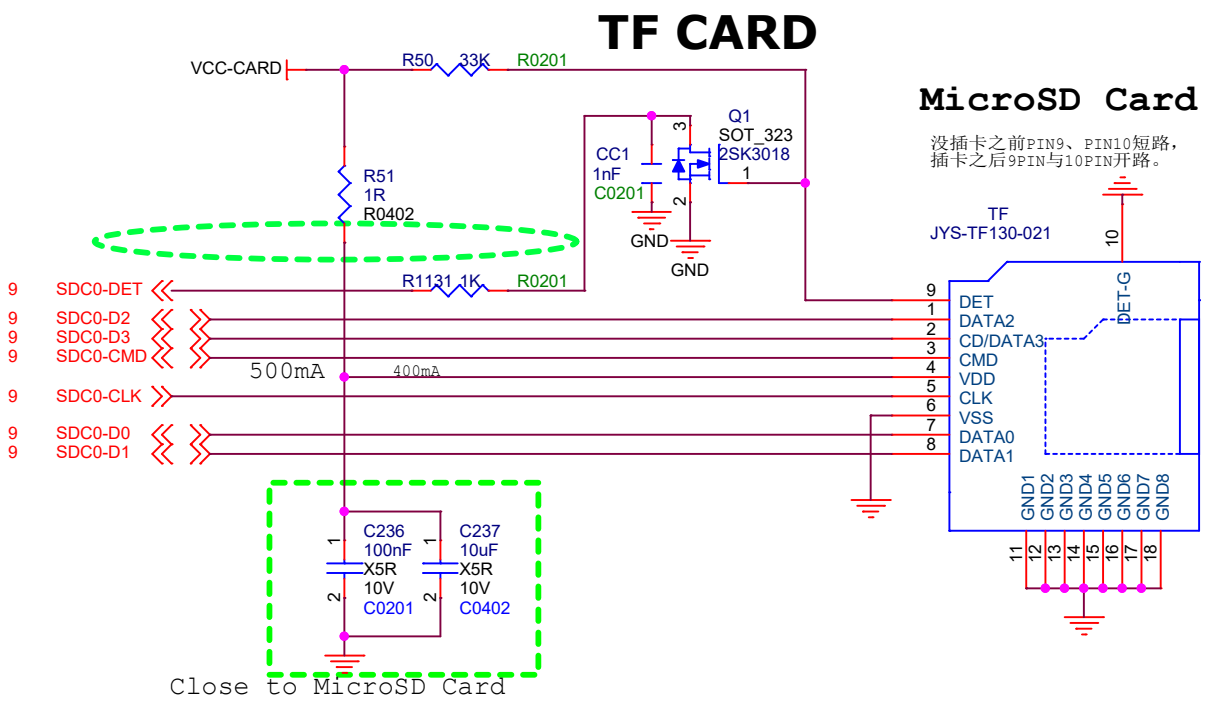


USB Type-C CC CTRL



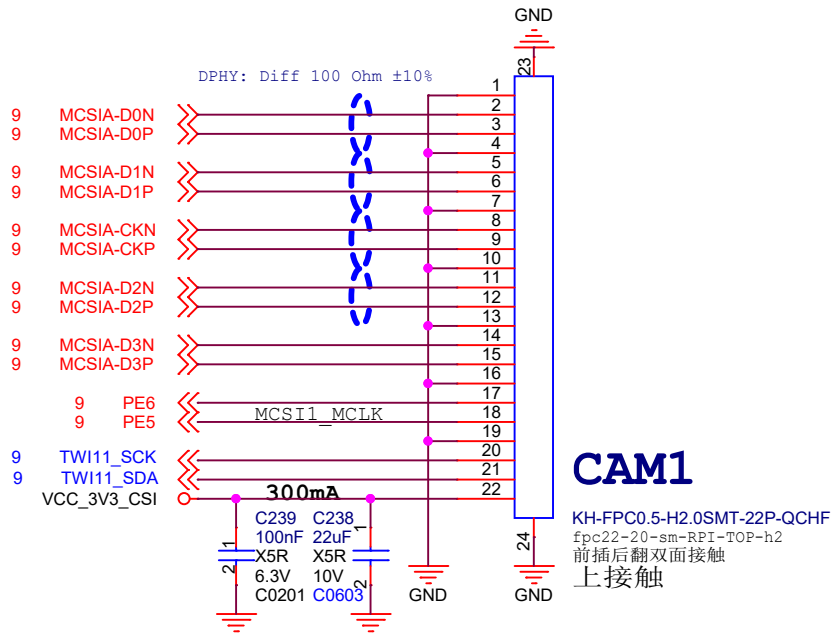
Shenzhen Xunlong Software Co., Ltd						
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Page Size:	A3	Sheet:	11	Of:	18	Rev: V1.2

TF CARD



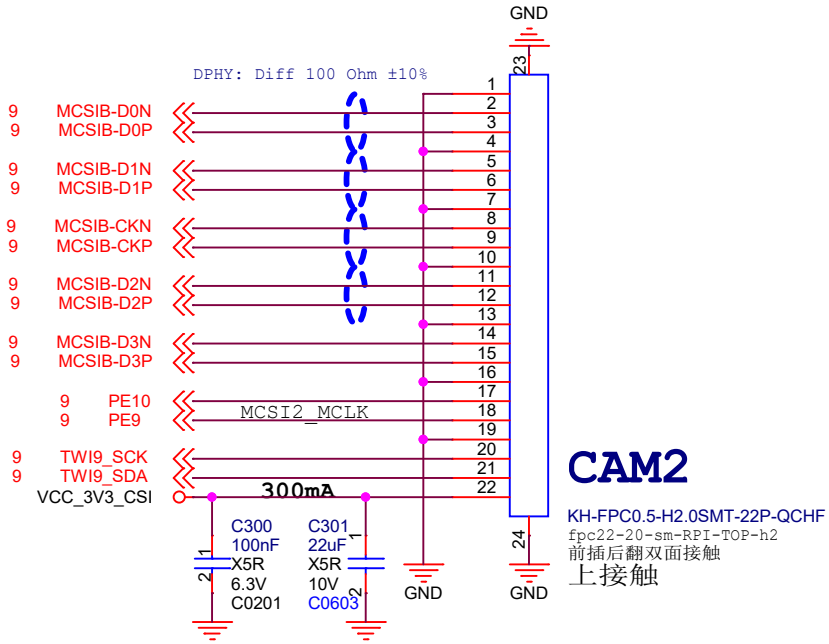
Shenzhen Xunlong Software Co., Ltd							
Schematic Name:	OPi ZERO 3W						
Page Name:	12_TF Card						
Page Size:	A3	Sheet:	12	Of:	18	Rev:	V1.2

MIPI_DPHY_CSI0



CAM1

KH-FPC0.5-H2.0SMT-22P-QCHF
fpc22-20-sm-RPI-TOP-h2
前插后翻双面接触
上接触



CAM2

KH-FPC0.5-H2.0SMT-22P-QCHF
fpc22-20-sm-RPI-TOP-h2
前插后翻双面接触
上接触

Shenzhen Xunlong Software Co., Ltd							
Schematic Name:	OPi ZERO 3W						
Page Name:	13_CAM_MIPI CSI						
Page Size:	A3	Sheet:	13	Of:	18	Rev:	V1.2

FCS960K

Note:VBAT peak-current is at least 500mA.

RF Microstrip
Z0= 50 ohm

VCCIO_1V8_WIFI

design for test

R0201 R0201 R0201 R0201 R0201 R0201

R54 R55 R56 R57 R58 R59

9 WL-SDIO-D2 — SDIO_D2_M0_WIFI
9 WL-SDIO-D3 — SDIO_D3_M0_WIFI
9 WL-SDIO-CMD — SDIO_CMD_M0_WIFI
9 WL-SDIO-CLK — SDIO_CLK_M0_WIFI
9 WL-SDIO-D0 — SDIO_D0_M0_WIFI
9 WL-SDIO-D1 — SDIO_D1_M0_WIFI

9 WL-REG-ON — PM1 — WL_PWR_EN
9 AP_WAKE_WL — PM2 — HOST_WAKE_WLAN
9 WL-WAKE-AP — PM0 — WIFI_WAKE_HOST

R0201 R1118 100K

9 AP-WAKE-BT — PM3 — AP_WAKE_BT
9 BT-WAKE-AP — PM4 — BT_WAKE_AP

9 I2S1_DIN0 — BT_PCM_DOUT
9 I2S1_BCLK — BT_PCM_CLK
9 I2S1-DOUT0 — BT_PCM_DIN
9 I2S1_LRCK — BT_PCM_SYNC

PCM_DIN/PCM_DOUT
与主控交叉连接

VCCIO_1V8_WIFI

R1119 10K R0201
R1120 10K R0201

RX/TX，CTS/RTS
主控端已经交叉

9 BT-UART-RX — BT_UART_RX
9 BT-UART-TX — BT_UART_TX
9 BT-UART-RTS — BT_UART_RTS
9 BT-UART-CTS — BT_UART_CTS

C274 220pF X5R 10V C0201

8 AP-CK32K-OUT

VCC-3V3-WIFI

UPZ1005D300-1R7TF
FB1 L0402

C240 10uF X5R 10V C0402
C241 220pF X5R 10V C0201
C243 100nF X5R 10V C0201

BT_WAKE_AP
AP_WAKE_BT

ANT
G1 G2 G3
IPEX-4
IPEX-4

U5 CDW-20800D8-00

WiFi6 11ax + BT5.4

XTAL_OUT XTAL_IN VBAT BT_VIO/N_HOST_WAKE BT_HOST_WAKE BT_WAKE N_WAKE FM_RX GND1 WL_BT_ANT GND

WL_PWR_EN L=OFF, H=ON
WIFI_WAKE_HOST
SDIO_D2_M0_WIFI
SDIO_D3_M0_WIFI
SDIO_CMD_M0_WIFI
SDIO_CLK_M0_WIFI
SDIO_D0_M0_WIFI
SDIO_D1_M0_WIFI
NC

WL_REG_ON
WL_HOST_WAKE
SDIO_DATA_2
SDIO_DATA_3
SDIO_DATA_CMD
SDIO_DATA_CLK
SDIO_DATA_0
SDIO_DATA_1
GND2
VIN_LDO_OUT
VDDIO

N_VDDSWP_IN
N_VDDSWP_OUT
N_VDDSWPIO

UART_CTS_N — BT_UART_CTS
UART_RXD — BT_UART_RX
UART_TXD — BT_UART_TX
UART_RTS_N — BT_UART_RTS
TX1
TX2
N_REG_PU
N_I2C_SCL
GND5
N_I2C_SDA
BT_RST_N

GPIOB6
NC
R61 0R R0201
HOST_WAKE_WLAN
兼容：BT_REG_ON

VIN_LDO LPO PCM_OUT PCM_CLK PCM_IN PCM_SYNC VDD_TCXO_IN GND3 GPS_RF GND4

NCNC NCNC NC

VCCIO_1V8_WIFI

C246 1uF X5R 10V C0201
C247 100nF X5R 10V C0201

20mA

AP-CK32K-OUT
R0201 0R/NC R1129
兼容设计
HOST_WAKE_WLAN
R1130 0R R0201

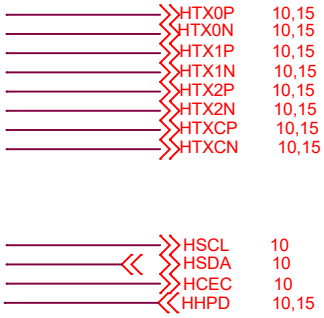
FCS960K

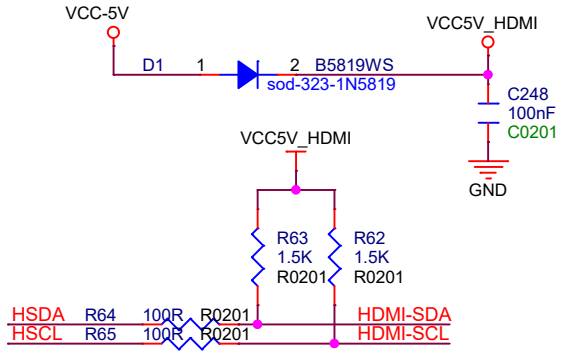
BT_PCM_SYNC
BT_PCM_DIN
BT_PCM_CLK
BT_PCM_DOUT

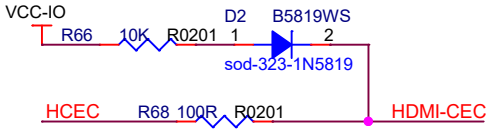
1. VBAT 不晚于 VDD_IO 上电。
2. VBAT 上电后至少 200 μs，PWRKEY 达到高电平。
3. PWRKEY 达到高电平后，至少再过 8 ms，模块完成开机。
4. PWRKEY 达到低电平后，至少再过 6 ms，模块完成关机。

Shenzhen Xunlong Software Co., Ltd						
Schematic Name:	OPi ZERO 3W					
Page Name:	14_WiFi/BT					
Page Size:	A3	Sheet:	14	Of:	18	Rev: V1.2

HDMI

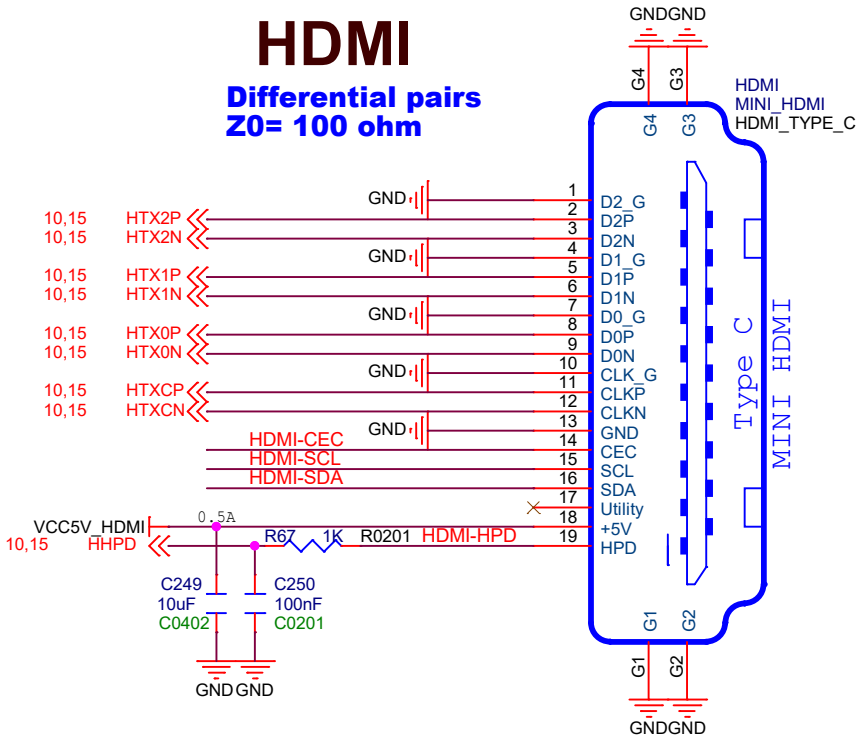




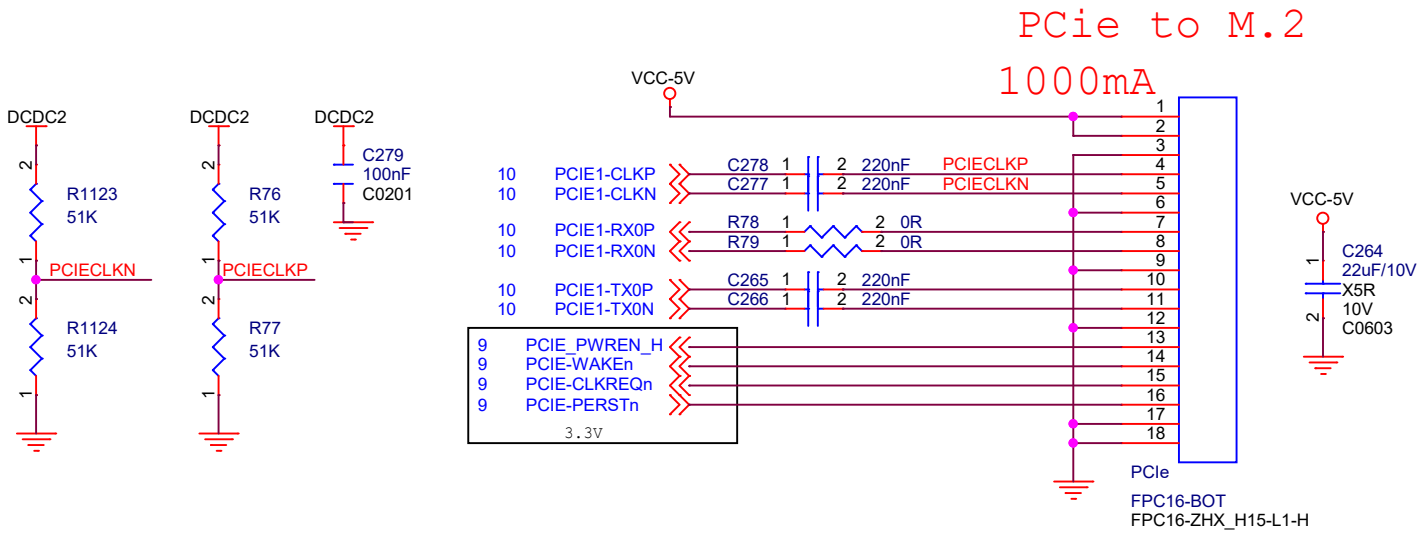


HDMI

Differential pairs
Z0= 100 ohm



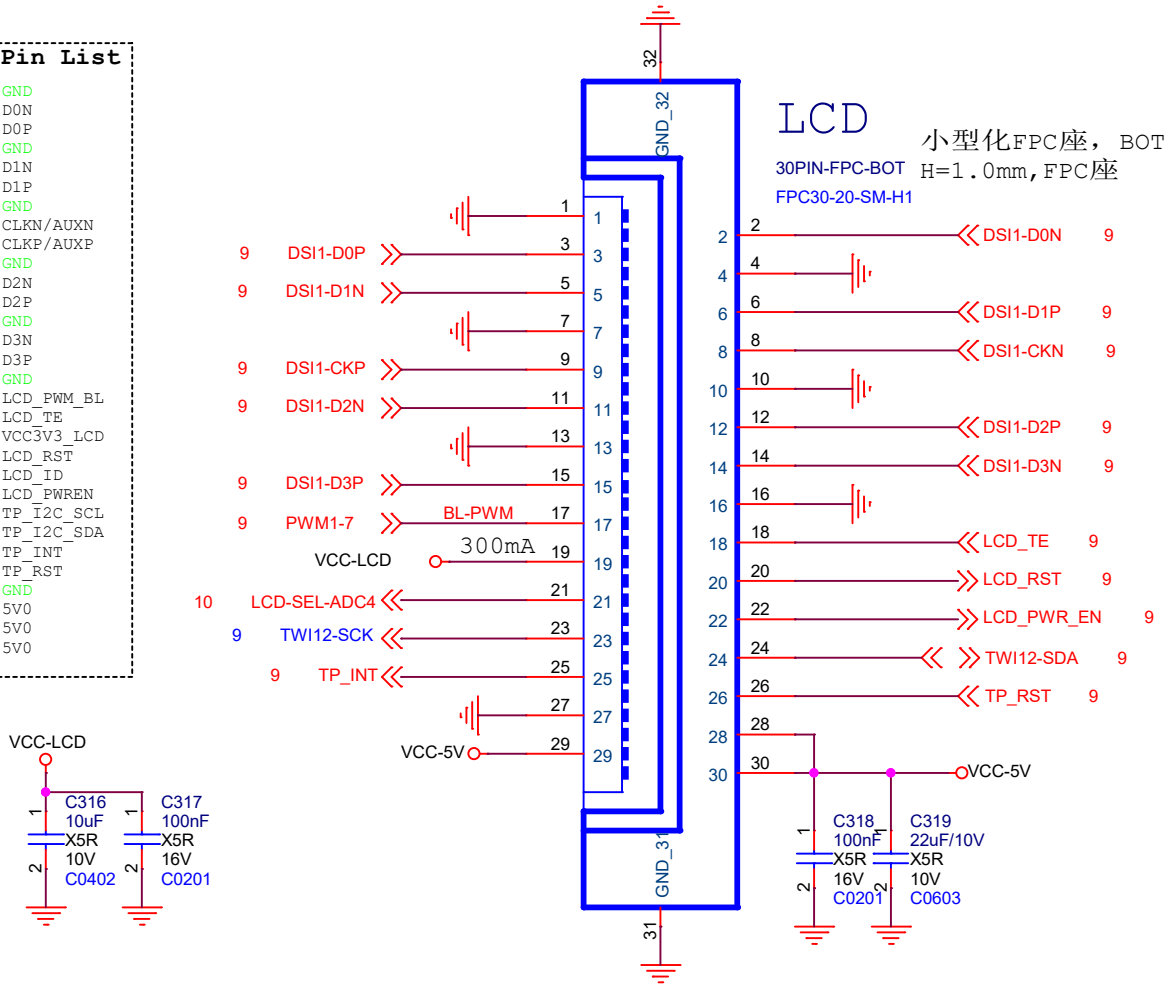
Shenzhen Xunlong Software Co., Ltd						
Schematic Name:	OPi ZERO 3W					
Page Name:	15_HDMI2.0					
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Schematic Name:	OPi ZERO 3W						
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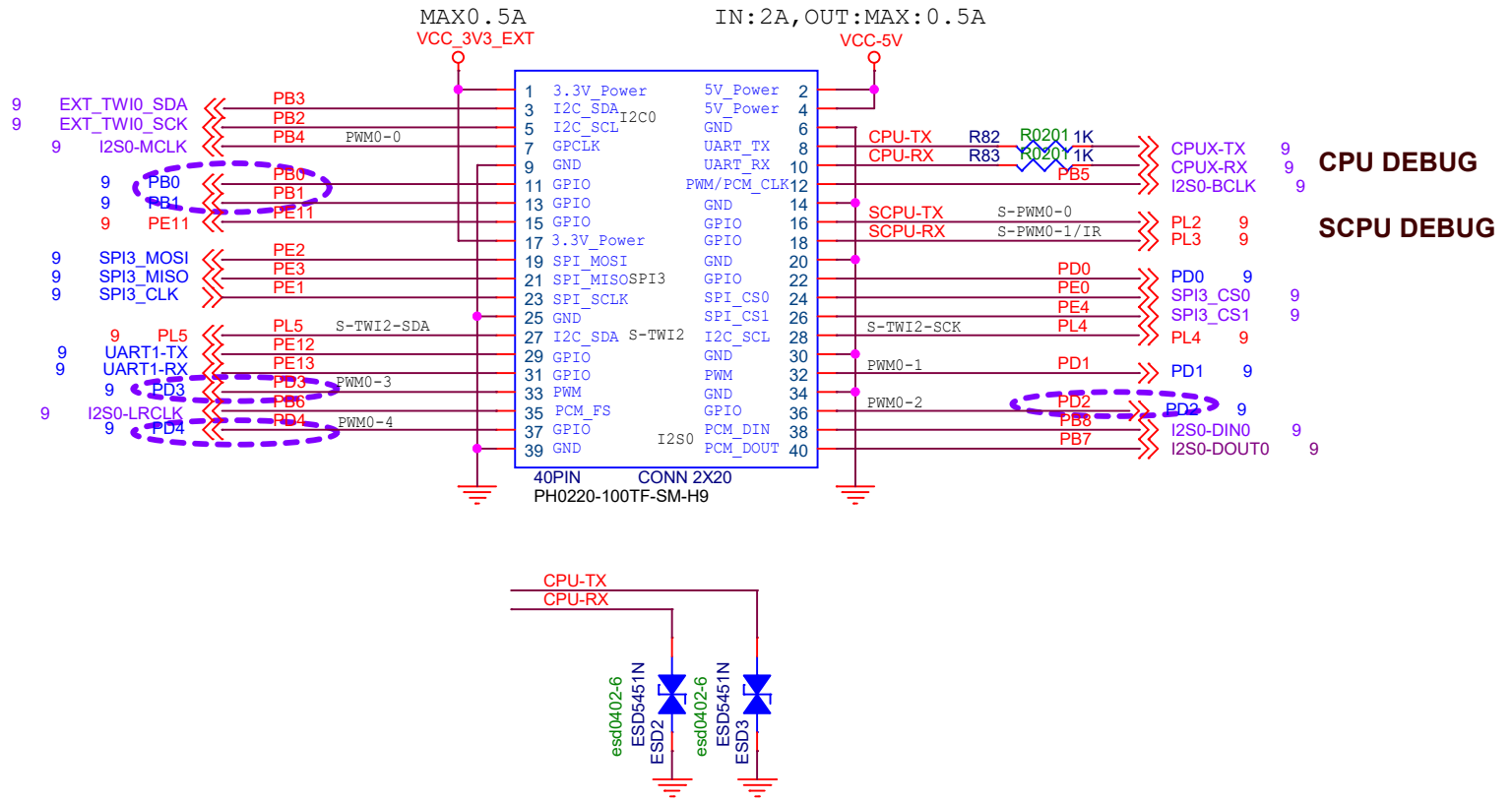
FPC Pin List

Pin1 :GND
Pin2 :D0N
Pin3 :D0P
Pin4 :GND
Pin5 :D1N
Pin6 :D1P
Pin7 :GND
Pin8 :CLKN/AUXN
Pin9 :CLKP/AUXP
Pin10:GND
Pin11:D2N
Pin12:D2P
Pin13:GND
Pin14:D3N
Pin15:D3P
Pin16:GND
Pin17:LCD_PWM_BL
Pin18:LCD_TE
Pin19:VCC3V3_LCD
Pin20:LCD_RST
Pin21:LCD_ID
Pin22:LCD_PWREN
Pin23:TP_I2C_SCL
Pin24:TP_I2C_SDA
Pin25:TP_INT
Pin26:TP_RST
Pin27:GND
Pin28:5V0
Pin29:5V0
Pin30:5V0



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EXT I/O



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